

APJ Abdul Kalam Technological University

MASTER OF PLANNING

URBAN PLANNING

CURRICULUM

A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Master of Planning ((Urban Planning) – Curriculum

JUNE 2025



A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Master of Planning (Urban Planning) – Curriculum

FIRST SEMESTER

Course Code	Course	L-T-S	Credits	Internal Marks	End Semester Exam		Exam Slot
					Marks	Duration (Hours)	
251TPL001	Urban Planning History and Theory	2-1-0	3	40	60	2.5	A
251TPL002	Socio Economic Base for Planning	2-1-0	3	40	60	2.5	B
251TPL003	Ecology, Environment & Disaster Resilience	2-1-0	3	40	60	2.5	C
251TPL004	Housing	2-1-0	3	40	60	2.5	D
251RPL001	Research Methodology and Planning techniques	2-1-0	3	40	60	2.5	S
251LPL001	GIS Applications I	0-2-0	2	40	-	-	-
251SPL001	Area Planning Studio	0-0-10	5	160	100	-	V*
	TOTAL	27	22	400	400		

*Jury Examination

SECOND SEMESTER

Course Code	Course	L-T-S	Credits	Internal Marks	End Semester Exam		Exam Slot
					Marks	Duration (Hours)	
252TPL001	Regional Planning and Resource Development	2-1-0	3	40	60	2.5	A
252TPL002	Planning and Environmental Legislation	1-1-0	2	40	60	2.5	B
252TPL003	Urban Economy and Employment	1-1-0	2	40	60	2.5	R
252TPL004	Urban Utilities and Infrastructure Planning	2-1-0	3	40	60	2.5	S
252EPL***	Interdisciplinary Elective	2-1-0	3	40	60	2.5	T
252EAR***	Department Elective	1-1-0	2	40	60	2.5	U
252LPL001	GIS applications II	0-2-0	2	40	-	-	-
252SPL001	Planning studio II	0-0-10	5	160	100	-	V*
		27	22	440	460		

*Jury Examination

Interdisciplinary Electives:

(Offered by Planning Stream for students in second semester of other branches)

252EPL001	Barrier free Built Environment
252EPL002	Circular Economy
252EPL003	Urban Analytics

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Department Electives: <i>(Offered to students of all M Arch and M Plan programmes)</i>	
252EAR001	Urban Conservation and Regeneration
252EAR002	Environmental Impact Assessment (EIA)
252EAR003	Urban Sociology

THIRD SEMESTER

Course Code	Course	L-T-S	Credits	Internal Marks	End Semester Exam		Exam Slot
					Marks	Duration (Hours)	
253TPL001	Urban Governance, Policies and Finance	2-1-0	3	40	60	2.5	A
253TPL002	Transportation planning	2-1-0	3	40	60	2.5	B
253TPL003	City and Metropolitan Planning	1-1-0	2	40	60	2.5	S
253EPL***	Program Elective	1-1-0	2	40	60	2.5	T
253PPL001	Dissertation	0-3-0	3	100	100	-	U*
253SPL001	Planning studio III	0-0-12	6	200	100	-	V*
253IAR001	Internship	-	2	100	-	-	W*
	TOTAL	25	21	560	440		

*Jury Examination

Program Electives:	
253EPL001	Planning for Tourism
253EPL002	Village planning and Rural Development
253EPL003	Inclusive Urban Planning
253EPL004	Real Estate Development
253EPL005	Disaster management

FOURTH SEMESTER

Course Code	Course	L-T-S	Credits	Internal Marks	End Semester Exam		Exam Slot
					Marks	Duration (Hours)	
254MPL001	MOOC Course	-	2	-	-	-	
254PPL001	Planning Thesis	0-0-26	12	350	150	-	V*
	TOTAL	26	14	350	150		

*Jury Examination

Note:

MOOCs must be from recognized institutes and approved by the faculty advisor as per regulations. Results must be available by the end of the fourth semester. Detailed instructions are provided in the syllabus.





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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251TPL001	Urban Planning History and Theory	2-1-0	3	2025
Course Objectives	• To understand economic, political, cultural, social and other forces shaping built environments in history • To understand the significance of theories in planning of settlements			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	To demonstrate appreciation and knowledge about history and theory of ancient cities' planning and development.	Understanding		
CO 2	To analyse various city planning concepts, physical forms and abstract theoretical formulations.	Understanding and Analyzing		
CO 3	To examine planning theories and approaches from modernist to post-modernist and participatory models.	Understanding and Analysing		
CO 4	To interpret the evolution and challenges of urban planning in India, with reference to historical and contemporary case studies.	Applying and Analysing		
CO 5	To critically evaluate emerging urban paradigms such as sustainable, smart, inclusive, and resilient cities in a global context.	Evaluating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The				

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questions in this part will carry 10 marks and can have sub questions.

COURSE PLAN			
Module	Contents	CO	Hours
I	Evolution of Planning – Planning history – Pre historic & Ancient Need to study history of planning and evolution of human settlements; ; Hunter, gatherer, farmer and formation of organised society; origins and growth of cities; Basic elements of the city; Historic determinants of settlement evolution: - geographical, climatic, socioeconomic, cultural, political, defensive, etc. Human settlements as an expression of civilizations; Ancient civilizations – Egypt, Mesopotamia, Greek	CO1	7
II	Planning history-Medieval to current – Medieval, Renaissance, Industrial and post industrial cities; Model and New Town Movements; Town Improvement and City Beautiful Movements Contribution of individuals to city planning: Patrick Geddes, Ebenezer Howard, Camillo Sitte, Lewis Mumford, Le Corbusier, C.A. Doxiadis, Clarence Perry, Peter Hall, William Alonso etc.	CO1, CO2	9
III	Planning Theories – Theory of city form: normative models – cosmic, machine, organic; Concentric Zone Theory, Sector Theory, Multiple Nuclei Theory; Modes of planning – blue print, synoptic, incremental, transactive, advocacy, participatory, mixed scanning; political economy approach - communicative model, new urbanism, and just city. Modernism and Post-Modernist Planning Thought; Neo-Marxist and Neo Liberal perspectives in Planning; Post positivist typology of planning theory, Comprehensive Rational Planning Approaches; Disjointed Incrementalism and Mixed Range Approach.	CO2, CO3	13
IV	Planning in India - History of urbanisation in India – ancient, medieval, pre	CO4	12

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	colonial, colonial, new towns; spatial patterns of urbanisation, settlement systems, classification of cities; Character of Indian cities and challenges involved in planning Post-independence Indian Planning and concept applications through case studies: Capital City (an amalgamation of cities) – Delhi; Chandigarh, Gandhi Nagar and Bhuvaneshwar, Steel Cities, Raipur; New Mumbai etc.		
V	Emerging concepts in planning – Global cities, 15 minute city, Sustainable Cities, Healthy City, Resilience City, Smart city, information city, inclusive city, safe city, shadow city, divided city, eco city etc.	CO5	7

References:

1. Hiraskar G.K (2012), “*Fundamentals of Town Planning*”, Dhanpat Rai Publications.
2. John Ratcliffe (1985), “*An Introduction to Town and Country Planning*”, Hutchinson.
3. Anthony James Catanese and James Synder C (1988), “*Introduction to Urban Planning*”, McGraw-Hill, Inc,US.
4. Davidoff.P,(1965): Advocacy and Pluralism in Planning, Journal of American Institute of Planners, Vol. 31. USA.
5. Amiya Kumar Das (2007), “*Urban Planning in India*”, Rawat Pubns.
6. Government of India (2015), “*Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines*, Vol I&II, Town and Country Planning Organization, Ministry of Urban Development, New Delhi.
7. Palermo, Pier Carlo (2010), “*Spatial Planning and Urban Development: Critical Perspective*”, Springer.
8. Kamal Uddin and Bhuiyan Monwar Alam (2023), “*Public Participation Process in Urban Planning: Evaluation Approaches of Fairness and Effectiveness Criteria of Planning Advisory Committees*”, Routledge.
9. Biswas Hiranmay (2012), “*Principles of Town Planning and Architecture*”, VAYU Education of India.
10. Peter Hall and Mark Tewdwr-Jones (2019), “*Urban and Regional Planning*”, Routledge; 6th edition.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251TPL002	Socio Economic Base for Planning	2-1-0	3	2025
Course Objectives	To introduce key sociological concepts and to provide insight towards knowing the sources of demographic data collection and methods of population projection for application in spatial planning To provide basic knowledge of economic concepts, tools and to understand the relevance of economic principles in urban planning and its application as well as estimating demand. To familiarize the sociological impacts of globalization and urbanization particularly in the Indian context, and also to understand inclusive urban development by expose to policies and case studies.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	To understand fundamental sociological concepts, assess the impact of social structures and institutions, evaluate issues related to social inequality, slums, and squatter settlements within the context of spatial planning.	Understanding		
CO 2	To apply and critically analyze demographic data and population characteristics. To learn how to use Census data and other knowledge. To familiarize various techniques of projection and forecasting towards decision-making and policy formulation relevance to spatial planning.	Applying		
CO 3	To understand key economic principles, evaluate economic indicators such as GDP, HDI, and inflation. To assess economic factors related to land use planning principles and also long-term investment and development strategies in planning practice.	Understanding		
CO 4	To apply economic concepts like scarcity, efficiency, and agglomeration within urban contexts. To utilize economic tools for the analysis and estimation of demand, pricing, and costs associated with urban services and amenities.	Applying		
CO 5	To analyze the sociological implications of globalization and urbanization. To assess challenges like	Understanding & applying		

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	gentrification and marginalization. To understand gender and equity concerns in urban development, as well as evaluate approaches and policies promoting inclusive cities in India and Global.	
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100 Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.		

COURSE PLAN			
Module	Contents	CO	Hours
I	Fundamentals of sociology for planning Terminology: Culture, Social Institutions, social organizations, Social Stratification, Social Change; Concept of Social Structure, Relationship between Social structure and spatial structure. Urban and Rural Sociology. Social aspects of planning for Human Settlements, Slums and squatter settlement-case studies.	CO1	9
II	Demographic data and population studies in Planning Sources of demographic data- sources of Population data: Census, NSS (National Sample Survey) CRS, SRS etc., Population size, composition and distribution in India. Population growth and transition in India, Population forecasts and projection methods, Population Policy in India Migration and its implication in settlement planning and development, case studies	CO2	9
III	Fundamental of Economics for Planning	CO3	11

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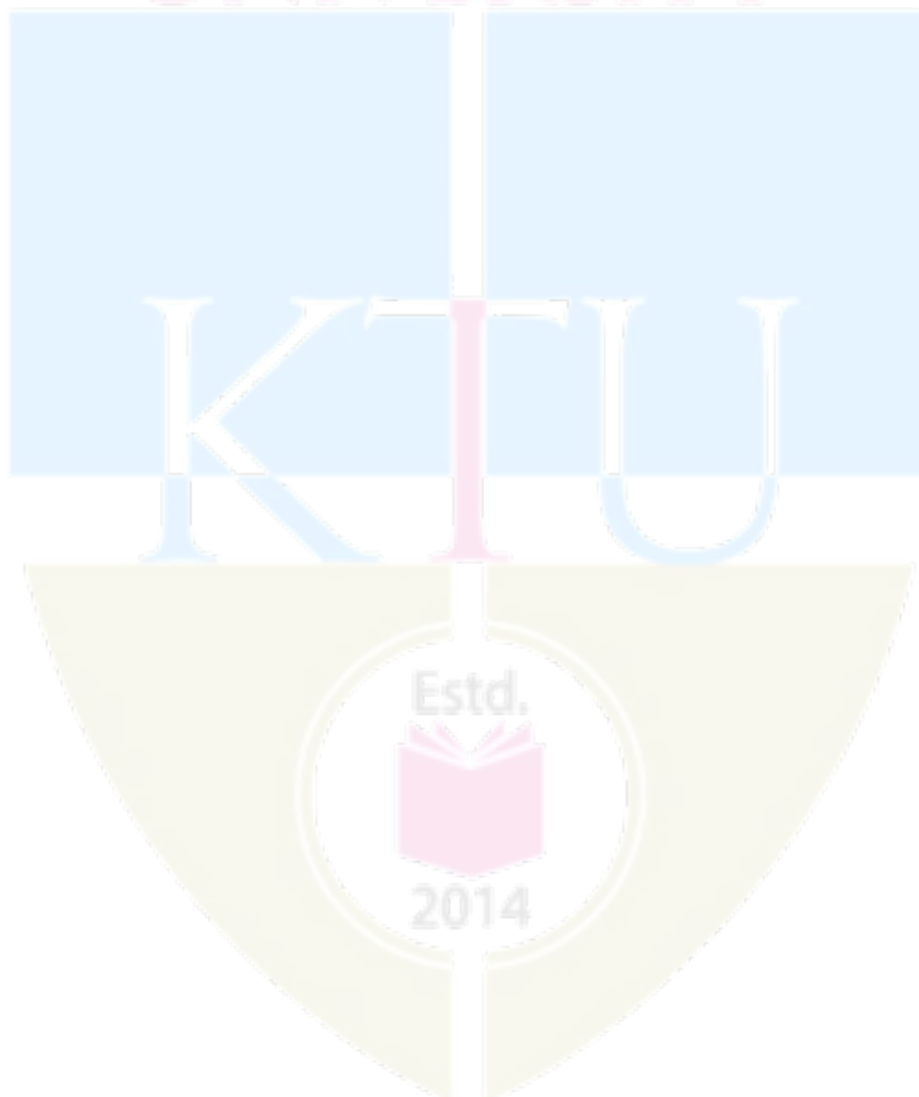
	Economic growth and Economic development, Human Development Index, GDP, GNP & Per capita Income, Concept of Demand & Supply, Economies of Scale; Micro Economics and Macro Economics, differences and examples, Production and consumption Production economics: Factors of production- land, labour, capital, entrepreneurship Land economics and Economic principles of land values and land use planning. Determinants of National Income, Inflation, Budgetary process in India, Short-term and Long-term investment planning.		
IV	Urban Economics and Planning Applications Basic axioms of Urban economics, Economics of urban agglomeration, urban finance. Importance of economics in planning, Core concepts of economics: Scarcity & efficiency. Application of economic tools in the estimation of Demand and supply, Cost of urban services and amenities. Taxation system in India	CO4	9
V	Globalization and Inclusive development Introduction and meaning of Globalization, Sociological impacts of Globalization, Merits and Demerits; Urbanisation and its impacts in the Indian Scenario: Gentrification, Ghettofication, Slumification, Marginalization, Gender and Equity concerns. National Commission on Urbanization; Public Participation and urban and rural development. Elements of Inclusive growth and Development. Inclusive development through local government and governance, Inclusive cities-case studies.	CO5	10

References:

1. Bhushan, Vidya & D.R. Sachdeva. *An Introduction to Sociology*, Kutab Mahal distributors, 2014.
2. Gispert P, (2015) “Fundamental of Sociology”, Orient Blackswan Pvt Limited Delhi.
3. Giddens, Anthony & Philip W. Sutton. *Sociology - Introductory Readings*, Polity, Oxford, 2010.
4. Wilson, R.A. & D.A. Schulz. *Urban Sociology*, Prentice Hall Inc., New Jersey, 1978.
5. United Nations. *Manual VIII, Methods for projections of Urban and Rural Population*, Department of Economics and Social Affairs, Population studies, No.55, New York, 1974.

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6. Sachithanandan, A.N., *Socio Economic Base for Planning – Teaching Material* prepared for the Institute of Town Planners, India.
7. McGann, Philip. *Modern Urban and Regional Economics*, Oxford University Press, London, 2013.
8. Sathish Y. Deodhar. *Day to Day Economics*, IIM Ahmedabad, 2012.
9. Smith, Wallace F., *Housing: The Social and Economic Elements*, University of California Press, 1971.
10. Chapin, F.S. *Urban Land Use Planning*, Higg & Brothers, New York, 1965.
11. Chen, Xiangming. *Introduction to Cities: How Places and Space Shape Human Experience*, Wiley Blackwell, UK., 2012.
12. Tonkins, Frank. *Cities by Design: The Social Life of Urban Form*, Polity Publications, UK. 2013.
13. Carter, H., *The Study of Urban Geography*, Edward A. Old, London, 1972.
14. *Global Review of Human Settlements*, Pergamon Press, London, 1976.
15. Kohun, Arthur, *History Builds the Towns*, Lund Humphries, London, 1953.



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251TPL003	Ecology, Environment & Disaster Resilience	2-1-0	3	2025
Course Objectives	- To introduce students to the fundamentals of ecology, environmental management and climate resilience - To introduce the fundamental principles of ecology and their relevance to urban environments and planning. - To understand key environmental challenges, including pollution, climate change, and their impacts on urban areas and infrastructure. - To examine disaster risks, vulnerability, and resilience strategies at the levels of settlements and housing. - To explore institutional frameworks, policies, and legal instruments for environmental management and disaster resilience.			
Course Outcomes: The course outcomes are as follows:				
CO	Description of Course Outcome	Learning level		
CO 1	To introduce fundamental ecological principles and their relevance to urban environments and planning.	Understanding and Analyzing		
CO 2	To explore urban environmental challenges, climate change impacts, and the planning responses necessary for resilience.	Analyzing and Evaluating		
CO 3	To understand disaster risk, vulnerability assessments, and resilience strategies at settlement and housing levels.	Understanding and Analyzing		
CO 4	To examine institutional frameworks, policies, and legal instruments guiding environmental management and disaster resilience.	Understanding and Analyzing		
CO 5	To analyse real-world examples and research-driven approaches to resilience, recovery, and sustainable planning.	Applying and Analyzing		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100 Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				

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End Semester Examination pattern:

There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

COURSE PLAN			
Module	Contents	CO	Hours
I	Fundamentals of Ecology and Urban Environmental Systems Key Ecological concepts: ecosystems, biodiversity, food chains, biogeochemical cycles, ecosystem services Urban ecosystems and urban metabolism: Land use ecology, energy flows in cities. Human environment interactions: Impact of human activity on earth systems, Urbanisation, Land use change, Environmental degradation. Critical ecological challenges: Loss of wetlands, green cover depletion, coastal vulnerability Concepts of sustainability: Natural resource management, ecological footprint, circular economy Global frameworks: Millennium Ecosystem Assessment, Sustainable Development Goals (SDGs).	CO1	9
II	Environmental Degradation, Climate change and Planning Responses Urban environmental issues: Air, water, soil, and noise pollution; waste management and its planning relevance Climate change and cities: Greenhouse gas emissions, sea-level rise, heat stress, extreme weather events Risk to housing and infrastructure: Urban flooding, thermal discomfort, coastal erosion Planning responses: Environmental Impact Assessment (EIA) and Strategic Environmental Assessment (SEA) Climate-resilient housing and land use planning strategies, Urban adaptation and mitigation approaches.	CO2	9
III	Disaster Risk and Resilience in Urban and Housing Contexts Types of disasters: Natural (floods, cyclones, earthquakes) and human-made (industrial, environmental hazards) Core concepts: Hazard, exposure, vulnerability, capacity, risk, resilience. Disaster risk in housing vs. city-wide systems.	CO3	9

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	Disaster management cycle: Preparedness, response, recovery, mitigation . Principles of “Build Back Better” in housing and infrastructure. Contextual focus: Disasters specific to India and Kerala Risk assessment approaches: Tools for housing and settlement vulnerability mapping Risk reduction strategies:Community-Based Disaster Risk Management (CBDRM), Nature-based solutions (green infrastructure, wetlands restoration) Integration of Disaster Risk Reduction (DRR) in master plans and housing schemes.		
IV	Environmental and Disaster Governance: Policies and Institutions Global Frameworks and Agreements: Sendai Framework for Disaster Risk Reduction, Paris Agreement, 2030 Agenda for SDGs, Kyoto Protocol, New Urban Agenda. Global Institutions: UNDRR, UN-Habitat, UNEP, IPCC World Bank – Global Facility for Disaster Reduction and Recovery (GFDRR) National Policies and Legislation: Environmental Acts: Environment Protection Act (1986), Air (1981) and Water (1974) Acts, Forest Conservation Act (1980), Biological Diversity Act (2002), Wetlands Rules (2017), Coastal Regulation Zone Notification (CRZ 2011/2019) Disaster Management Frameworks: Disaster Management Act (2005), Environment Impact Assessment (EIA) Notification (2006), National Policy on Disaster Management (2009), EIA Notification (2006), State Disaster Management Plans (SDMPs) Governance Institutions: National: National Disaster Management Authority (NDMA), Ministry of Environment, Forest & Climate Change (MoEFCC), Ministry of Housing and Urban Affairs (MoHUA), Ministry of Home Affairs (MHA) State: State Disaster Management Authorities (SDMAs), State Pollution Control Boards (SPCBs), Town & Country Planning Departments District/Local: Disaster Management Authorities (DDMAs), Urban Local Bodies (ULBs), Municipal Corporations, Panchayats.	CO4	12

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V	Case Studies in Sustainable and Resilient Urban Planning Post-tsunami reconstruction in Tamil Nadu Resilient housing in Bangladesh floodplains Kerala floods (2018) – response and recovery Cyclone resilience planning in Odisha Eco-districts in Portland, USA Netherlands: Flood-resilient cities and living with water strategies Copenhagen: Climate-resilient neighbourhoods Surat: Urban Climate Resilience Strategy.	CO5	9
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References:

1. Agarwal, S. K. (2011). *Fundamentals of ecology*. APH Publishing Corporation.
2. Alberti, M. (2008). *Advances in urban ecology: Integrating humans and ecological processes in urban ecosystems*. Springer.
3. Foreman, R. T. T. (2014). *Urban ecology: Science of cities*. Cambridge University Press.
4. Hoonweg, D. (2011). *Cities and climate change: Responding to an urban agenda*. World Bank Publications.
5. Jensen, J. R. (2007). *Remote sensing of the environment: An Earth resource perspective*. Pearson.
6. Ministry of Housing and Urban Poverty Alleviation, Government of India. (2016). *Disaster risk reduction: A handbook for urban managers*.
7. Niemelä, J., Breuste, J. H., Guntenspergen, G., McIntyre, N. E., Elmqvist, T., & James, P. (Eds.). (2011). *Urban ecology: Patterns, processes and applications*. Oxford University Press.
8. Odum, E. P., & Barrett, G. W. (2005). *Fundamentals of ecology* (5th ed.). Cengage Learning.
9. Saksena, K. D. (1993). *Environmental planning, policies and programmes in India*. Shipra Publications.
10. Sethi, M. (2017). *Climate change and urban settlements: A spatial perspective of carbon footprint and beyond*. Routledge.
11. Shaw, R., & Krishnamurthy, R. R. (2009). *Disaster management: Global challenges and local solutions*. University Press.
12. Wisner, B., Blaikie, P., Cannon, T., & Davis, I. (2004). *At risk: Natural hazards, people's vulnerability and disasters* (2nd ed.). Routledge.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251TPL004	Housing	2-1-0	3	2025
Course Objectives	• This course introduces the fundamental terminology, concepts, and socio-economic dimensions of housing. • The primary objective of this course is to enhance students understanding of global housing challenges and underscore the importance of housing and related infrastructure at both the individual and national levels. • Students will be introduced to the various approaches, issues, and considerations of housing.			
Course Outcomes: After the completion of the course, the students shall be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand global challenges and the importance of housing in developing nations.	Understanding		
CO 2	Understand the socio-economic dimension of housing	Understanding		
CO 3	Evaluate the housing needs and develop a high-quality living environment.	Analysing		
CO 4	Investigate the different approaches and norms in housing.	Analysing		
CO 5	Analyse the case studies for various approaches in housing and precedents derived from innovative housing initiatives.	Analyzing and Applying		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern:				

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There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

Module	Contents	CO	Hours
I	Concepts and Definitions The concept of housing, its relevance, shelter as a basic requirement- Maslow's hierarchy of needs, and the Determinants of housing. Role of housing in social and economic development Census Definitions: Housing Stock, Housing Shortage, Housing Need, and Demand. Housing density- net density & gross density, obsolescence, occupancy rate Housing Situations in developing and developed countries: understand the housing scenario in the global context. Housing Situations in India: The Existing housing stock in India. International declarations of Housing – SDG, MDG, Habitat Agenda, IYSH, New Urban Agenda, Niti Aayog. National Housing Policy, National Urban Housing & Habitat Policy (2007), National Rural Housing & Habitat Policy.	CO1	9
II	Social-Economic Dimensions and Housing Market Housing and social security: the role of housing in community well-being; gender dimensions of housing; housing for the elderly. Economic dimensions: Contribution of housing to micro and macro economy, contribution to national wealth and GDP, national budgets, employment generation, affordability Housing market -Factors affecting local housing market, Housing bubble, Housing stress- Rental Housing in India	CO2	9
III	Norms in Housing and Housing Projections Housing Norms and Standards: National Building Code, URDPFI Guidelines, CPHEEO (MoUD) manual, Concept of Habitable Room, Habitable Dwelling Unit. Zoning -Understanding the zoning of housing in the city's development plan, the Location of residential zoning concerning other land uses.	CO3	10

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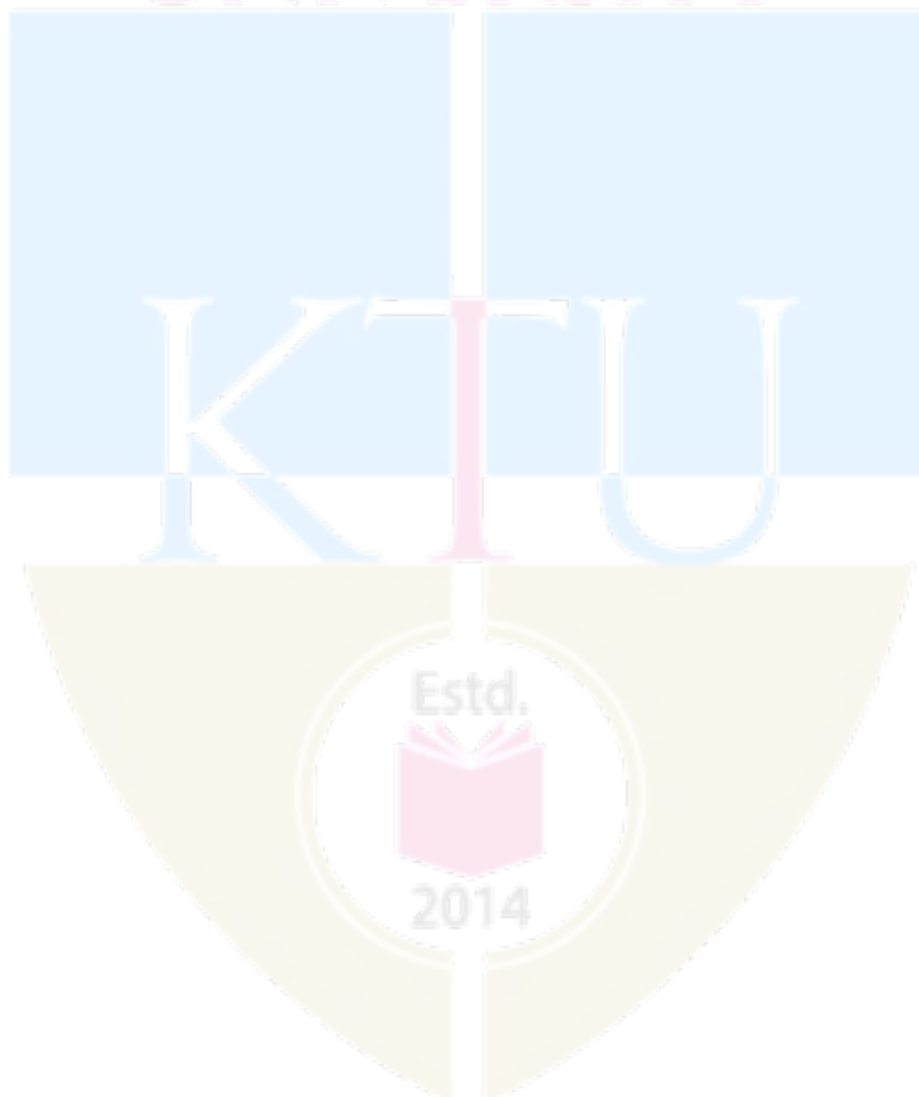
	Housing Projections- Estimating future housing requirements, examining land use provisions, and land suitability for housing development.		
IV	Housing Environment and Finance Housing environment: Housing for the poor, Slums and squatters, informal sector housing, segregation in housing, infrastructure, services, and facilities in housing areas, Public Health Issues related to housing, Residential satisfaction & its factors Housing finance- importance of housing finance, Major housing finance agencies at the international and national levels.	CO3 , CO4	9
V	Housing typologies and approaches to shelter development Typologies -Based on the characteristics of construction structural conditions, materials of construction, housing age, and density (low and high-rise), plotted development, group housing Approaches to shelter development: Settlement upgradation, incremental development, Site and services scheme, Core housing schemes, Self-help schemes, etc. Affordable Housing, Public Private Participation, Community participation in design and implementation of housing programs, Green Housing, Sustainable housing, Green Ratings (LEEDS, GRIHA, etc.). Analysis of case studies of different housing approaches	CO4 , CO5	11

References:

1. Abrams, C. (1964). *Man's struggle for shelter in an urbanising world*. MIT Press/Harvard University.
2. Allen, W. A., Happold, E., Word, A. M., & Courtney (Eds.). (1992). *A global strategy for housing in the third millennium*. E & FN Spon.
3. Bardhan, A., Edelstein, R. H., & Kroll, C. A. (2011). *Global housing markets: Crises, policies, and institutions*. Wiley.
4. Beyer, G. H. (1958). *Housing: A factual analysis*. The Macmillan Company.
5. Farr, D. (2008). *Sustainable urbanism: Urban design with nature*. John Wiley & Sons.
6. Gallion, A. (1953). *Urban pattern*. Van Nostrand Reinhold.
7. Jain, A. K. (2009). *Urban housing and slums*. Bookwell Publications.
8. Kilmartin, L., & Singh, H. (Eds.). (1992). *Housing in the Third World: Analyses and solutions*. Oxford & IBH Publishing Co.
9. McCabe, B. J., & Rosen, E. (Eds.). (2023). *The sociology of housing: How homes shape our social lives*. Routledge.

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10. Payne, G. (1977). *Urban housing in the Third World*. Routledge and Kegan Paul.
11. Rapoport, A. (1969). *House form and culture*. Prentice Hall.
12. Ram, P., & Harper, M. (2020). *The affordable housing market in India: Institutional constraints, informal sector and privatisation*. Taylor & Francis.
13. Revi, A. (1990). *Shelter in India – Sustainable development series*. Stosius Inc./Advent Books Division.
14. Bureau of Indian Standards. (2005). *National Building Code of India*. New Delhi: BIS.
15. Ministry of Housing and Urban Affairs. (2015). *Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines*.
16. Government of Kerala. (2019). *Kerala Municipality Building Rules*.
17. United Nations. (1996, June 14). *The Habitat Agenda: United Nations Conference on Human Settlements (Habitat II)*. Istanbul.



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251RPL001	Research Methodology and Planning Techniques	2-1-0	3	2025
Course Objectives	The course attempts to : - To introduce various research paradigms, to understand and formulate a good research design. To learn about ethics in research and understand various data collection methods. To introduce students to various statistical methods in research. - To introduce the key concepts in planning techniques and newer techniques in planning. To learn about the data base for physical survey and understand the detailing for various levels of plans.			
Course Outcomes: After the completion of the course, the students shall be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand the concept of research in planning and analyze various processes involved in research design.	Understanding and Analyzing		
CO 2	To understand the concept of ethics in research. To understand different types of data, its measurement and analysis	Understanding and Analyzing		
CO 3	To apply statistical methods in research. To understand population forecasting and projections.	Applying		
CO 4	To understand the key concepts in planning techniques, To learn about databases for physical surveys and understanding the detailing for various levels of plans.	Understanding		
CO 5	To understand Techniques in settlement planning.	Understanding and Analyzing		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				

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End Semester Examination pattern:

There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

COURSE PLAN			
Module	Contents	CO	Hours
I	Introduction to research: Key Concepts, Constructs and Operational Definitions: Variables, Propositions, Hypothesis, Types of research. Research process: Literature review, Literature map, Identifying gap areas from literature review, Problem identification, Development of working hypothesis, Formulating research, Referencing styles. Research Design: Components of research design, Features of a good design, Qualitative and quantitative research designs, Examples of research designs in relevant fields- Architecture, Planning, Housing and Urban Design. Preparation of Research proposals and reports, Dissertation, Thesis, Scholarly articles.	CO1	10
II	Ethics in Research, plagiarism, Intellectual Property rights. Methods of data collection and analysis: Documents, Observation, Surveys, Experiments. Methods of data collection and analysis: Methods of data collection – Documents, Observation, Surveys, Experiment, Types of data and measurement, Data collection and analysis, Grounded theory, Ethnography, Phenomenology. Types of Analysis, Interpretation and Generalization.	CO2	8
III	Need for Statistical methods in research, Types of sampling, Sample size determination, Statistical Analysis, Descriptive, Inferential and Predictive statistics, Data distribution, Measures of central tendency- Mean, Median, Mode, Variance, Standard deviation and Data dispersion. Testing of Hypothesis, Statistical errors in Hypothesis testing, Anova, Chi Square tests, Correlation and Regression. Their application in planning.	CO3	11

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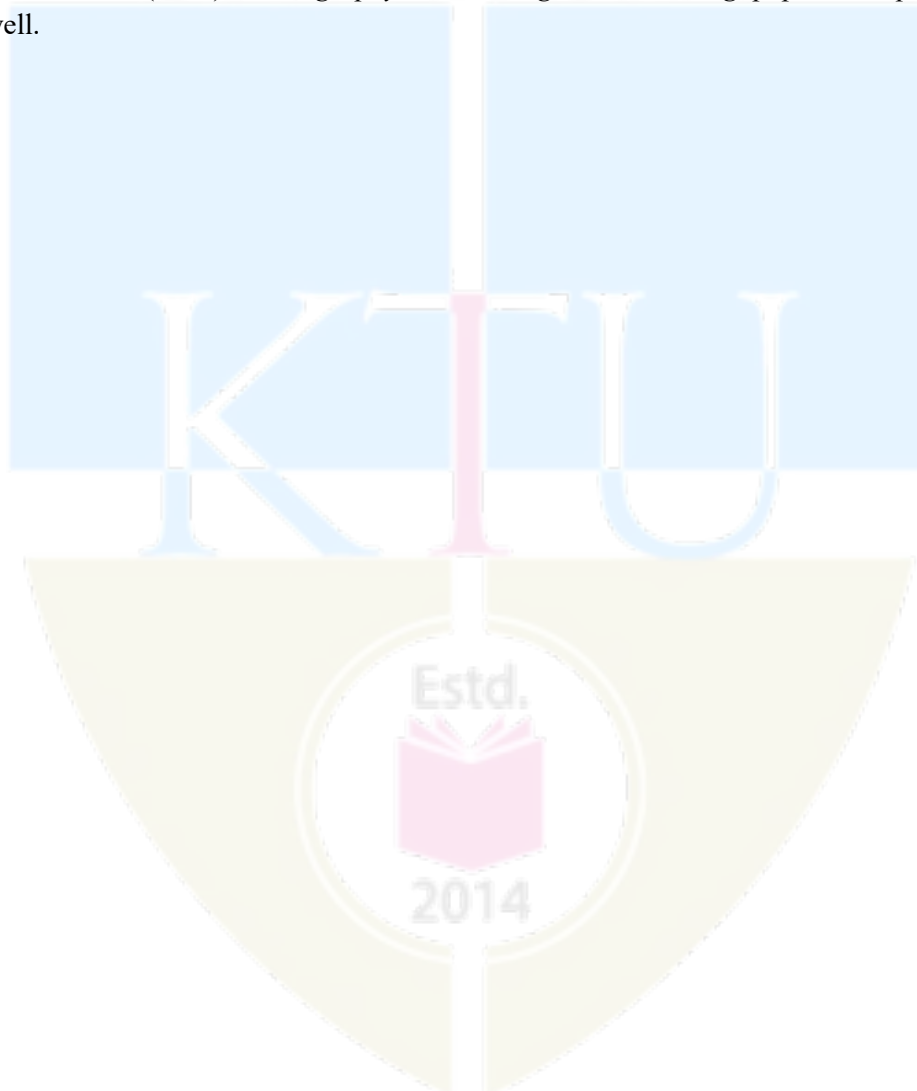
	Demography: Methods of population forecasts and projections techniques including Arithmetical increase, Geometrical increase, Incremental increase, Graphical method and Comparative Graphical method. Data analysis using Statistical softwares. (To be handled by faculty from Mathematics)		
IV	Planning Techniques: Introduction to planning techniques. Database for physical surveys including land use, building use, density, building age, etc., and socio-economic surveys; Sampling and survey techniques; Land use classification or coding and expected outputs. Understanding the concepts of scales, components and detailing for various levels of plans like Perspective plan, Regional plan, Development plan, Local area plan, special purpose plans and Annual plan. Land information systems – Aerial photography, remote sensing, photogrammetry, photo interpretation and mapping. Ranking of settlements; Guttman's Scalogram, Desire line diagrams, Threshold analysis: Input output analysis, SWOT analysis, Planning models (descriptive and decision making models) etc.	CO4	10
V	Techniques in settlement planning: Land pooling, land assembly, PRT (Plot reconstitution techniques), land readjustment, Transfer of Development Right, Various approaches to urban land zoning (mixed zone, floating zone, white zoning etc. TOD (Transit Oriented Development), PIU (Principles of Intelligent Urbanism), Public participation, Monitoring and evaluation in the planning process. Collaborative, Advocacy, inclusive and conflict resolution. Newer techniques in planning – through national/ international case studies.	CO5	9

References:

1. Babbie, E. (1983). *The practice of social research* (3rd ed.). Wadsworth Publishing Co.
2. Braken, I. *Urban planning methods*. Routledge.
3. Creswell, J. W. (1994). *Research design: Qualitative approaches*. Sage Publications.
4. Creswell, J. W. (2003). *Research design: Qualitative, quantitative, and mixed methods approaches* (2nd ed.). Sage Publications.
5. Day, R. A. (1989). *How to write and publish a scientific paper*. Cambridge University Press.

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6. De Vaus, D. A. (2002). *Surveys in social research*. Rawat Publications.
7. Gawatre, D. W., Kandgule, M. H., Kharat, S. D., et al. *Comparative study of population forecasting methods*.
8. Keeble, L. B. *Principles and practice of town and country planning*. Estates Gazette Ltd.
9. Kothari, C. R. (1990). *Research methodology*. Sultan Chand & Sons.
10. Kruekeberg, D. A., & Silvers, A. L. *Urban planning analysis: Methods and models*. John Wiley & Sons.
11. Majumdar, P. K. (2010). *Fundamentals of demography*.
12. Marko, & Mooi. (2019). *A concise guide to market research: The process, data and methods using IBM SPSS Statistics*.
13. NPTEL IIT Kharagpur. *Population forecasting* (Web course, Module 5, Lecture 5).
14. Robert, M. *An introduction to town planning techniques*. Hutchinson Educational.
15. Preston, S., et al. (2000). *Demography: Measuring and modeling population processes*. Wiley-Blackwell.



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CURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251LPL001	GIS Applications I	0-2-0	2	2025
Course Objectives	This course aims to enable students to learn and apply GIS during their academics and equip them to become proficient in the professional practice of applying GIS.			
Course Outcomes: After the completion of the course, the students shall be able to:				
CO	Description of Course Outcome		Learning level	
CO 1	To introduce the fundamentals of Geographic Information Systems (GIS) and its role in urban planning, including coordinate systems, data types, projections, and spatial referencing methods.		Understanding and Remembering	
CO 2	To develop skills in managing spatial data and performing thematic analysis using GIS software, including tasks such as querying, buffering, overlay analysis, and raster operations.		Understanding and Applying	
CO 3	To equip students with practical experience in using open-source GIS tools and conducting spatial analysis related to urban planning themes, such as land use, infrastructure, and environmental factors.		Applying	
CO 4	To foster a critical understanding of the ethical and professional issues in GIS-based planning practice, including data accuracy, and decision-support systems for sustainable development.		Applying	
Mark Distribution: Continuous internal evaluation - 40 marks				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 20 marks Internal test - 20 marks				

Module	Contents	CO	Hours
I	Introduction to GIS & Urban Data - GIS basics: Interface, data models (vector/raster), - Coordinate systems and their projections - Urban data sources (census, land use, infrastructure networks).	CO1, CO2	14

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	• Ethical considerations (data privacy, biases in urban data). • Navigate GIS Software and explore urban datasets (e.g., OpenStreetMap, city open-data portals). • Basemap preparation using GIS Software		
II	Spatial Analysis for Urban Planning • Spatial queries (zoning compliance, service area delineation). • Density analysis (population, employment, amenities). • Proximity analysis (schools, hospitals, transit stops). • Map walkability using buffer analysis. • Create a land-use suitability map using overlay techniques. • Principles and techniques of digital image classification- Supervised and Unsupervised classification (LULC Classification) • Download and process satellite imagery, perform Urban Heat Island (UHI) analysis using land surface temperature extraction (Landsat, Sentinel, and MODIS) • Python and other scripting possibilities for GIS	CO1, CO2, CO3, CO4	18

References:

1. Chang, K. T. (2019). *Introduction to geographic information systems* (9th ed.). McGraw-Hill Education.
2. Jensen, J. R. (2016). *Remote sensing of the environment: An earth resource perspective* (3rd ed.). Pearson.
3. Lo, C. P., & Yeung, A. K. W. (2007). *Concepts and techniques of geographic information systems* (2nd ed.). Pearson.
4. Lillesand, T. M., Kiefer, R. W., & Chipman, J. W. (2020). *Remote sensing and image interpretation* (8th ed.). Wiley.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251SPL001	Area Planning Studio	0-0-10	5	2025
Course Objectives	To comprehend the regional connectivity of the study area. - To comprehensively analyze the area and integrate infrastructure and utilities under current conditions. - The goal is to create all-encompassing solutions for physical, social, economic, and environmental infrastructure that comply with relevant planning guidelines and standards.			
Syllabus	The literature review is within disciplines associated with planning, site planning, and urban and rural areas. It aims to develop map preparation, report writing, and presentation skills.			
Course Outcomes: After the completion of the course, the students shall be able to:				
CO	Description of Course Outcome		Learning level	
CO 1	Critically analyse relevant literature and field conditions		Understanding and Analyzing	
CO 2	Apply planning techniques effectively to adhere to planning norms and standards.		Applying	
CO 3	Develop the ability to devise solutions for preparing and executing plans involving complex and interrelated factors within spatial planning.		Applying	
Mark Distribution: Continuous internal evaluation -160 (+ 40 marks for GIS Lab) Final Jury Evaluation (Viva) - 100 Total Marks - 260 (300 marks including GIS Lab)				

Module	Contents	CO	Hours
I	Local Area Planning Exercise Prepare an area plan for a part of a city with multiple uses/	CO1, CO2, CO3	160

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	housing areas/ urban-rural interfaces, etc. Literature review – This project begins with a literature review of relevant topics and areas from research papers, book chapters, journals, documents, reports, and other scholarly sources. The basic aim of this project is to understand rural and urban character through an area (layout/zone/part of settlement) study. The project entails a comprehensive planning approach that employs established norms and standards to enhance students' understanding of the fundamental principles of area planning/settlement planning, as derived from theoretical studies conducted throughout the semester. This process is designed to facilitate a multidisciplinary approach in addressing the various factors, utilities, and infrastructure that must be considered in spatial planning to attain balanced development. The relevant policies, strategies, and other considerations must be submitted as the outcome of this exercise. The project aims to enhance student engagement through various means, including data collection, fieldwork, stakeholder interaction, collaboration, critical analysis, survey methods, planning techniques, analytical tools, software applications, and presentation of study, encompassing map preparation and reporting compilation.		
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References:

1. Southall, A. (1998). *The city in time and space*. Cambridge University Press.
2. Perdicoulis, A. (2011). *Building competences for spatial planners: Methods and techniques for performing tasks with efficiency*. Taylor & Francis.
3. Alexander, E. R. (1992). *Approaches to planning: Introducing current planning theories, concepts and issues*. Gordon and Breach Science Publishers.
4. LaGro, J. A., Jr. (2013). *Site analysis: Informing context-sensitive and sustainable site planning and design*. John Wiley & Sons.
5. Jacobs, J. (1989). *The death and life of great American cities*. Knopf Doubleday Publishing Group.
6. Hall, P., & Tewdwr-Jones, M. (2010). *Urban & regional planning*. Routledge.
7. Ramachandran, R. (1991). *Urbanization and urban systems in India*. Oxford University Press.
8. Ministry of Urban Affairs & Employment, Government of India. (2014). *Urban and regional development plans formulation & implementation guidelines*. Government of India.



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252TPL001	Regional Planning and Resource Development	2-1-0	3	2025
Course Objectives	The course attempts to : - To introduce students to the discipline of regional planning, its evolution in India and world - To expose students to regional planning theories - To make students aware of the institutional mechanisms involved in regional planning and resource development			
Course Outcomes: The course objectives are as follows:				
CO	Description of Course Outcome	Learning level		
CO 1	To introduce key concepts, relevance, and methods of regional planning and delineation.	Understanding and Analyzing		
CO 2	To understand and apply theories of regional interaction, growth, and location.	Understanding and Applying		
CO 3	To examine the institutional and legislative frameworks for regional planning in India.	Understanding and Evaluating		
CO 4	To explore approaches for achieving balanced regional development and reducing disparities.	Analysing and Evaluating		
CO 5	To understand the role of regional planning in resource development and infrastructure.	Applying and Analysing		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.				

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COURSE PLAN			
Module	Contents	CO	Hours
I	Introduction to Regional Planning Introduction to regional planning: Aims, objectives, relevance, and scope Concepts and classification of regions Characteristics of a planning region Delineation of regions and regional delineation techniques Case studies in regional delineation: Factor analysis, Cluster analysis, Flow analysis Overview of regions in India UNCRD and regional development goals	CO1	10
II	Theories of Regional Interaction and Growth Regional interaction theories: • Central Place Theory • Zipf's Rank-Size Rule Regional growth theories: • Sector Theory • Export Base Theory • Input-Output Model Least-cost location theories: • Von Thunen's model • Weber's Industrial Location Theory	CO2	10
III	Institutional and Legislative Framework Institutional mechanisms for regional planning in India Role of 73rd and 74th Constitution Amendment Acts District as a region – scope and limitations Preparation and implementation of regional plans Planning for metropolitan regions: Examples from NCR, Mumbai, Chennai.	CO3	9
IV	Balanced Regional Development Role of regional planning in addressing regional imbalances Characteristics and reasons for backwardness Theories of regional economic divergence: • Cumulative Causation Theory • Growth Pole Theory	CO4	9

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	Misra's Growth Foci Approach		
V	Relevance of regional planning in resource development Resource regions in India: - Western Ghats - South-East Resource Region - Coastal and River Valley Regions - Industrial and Coastal Regions Infrastructure planning in regional development Planning for special areas and mega projects Case studies from other nations	CO5	10

References:

1. Chaudhuri, J. R., & Chaudhuri, R. (2001). An introduction to development and regional planning: With special reference to India. Orient Blackswan.
2. Glasson, J., & Marshall, T. (2007). Regional planning (2nd ed.). Routledge.
3. Hall, P., & Tewdwr-Jones, M. (2010). Urban and regional planning (5th ed.). Routledge.
4. Ministry of Urban Affairs & Employment, Government of India. (2014). Urban and regional development plans formulation & implementation guidelines. Ministry of Urban Affairs & Employment.
5. Wang, X., & von Hofe, R. (2008). Research methods in urban and regional planning. Springer Science & Business Media.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252TPL002	Planning and Environmental Legislation	1-1-0	2	2025
Course Objectives	The course attempts to, - To understand the relevance of constitution and legislation in relation to spatial planning - To make aware of the importance of spatial and environmental planning laws, legislations, acts and regulations in India			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand terminologies and the relevance of legislation studies to the planning profession.	Understanding		
CO 2	To acquire foundational knowledge of legal concepts, the Indian constitution, and specific acts, laws, rules, and regulations related to Environment and urban planning.	Understanding		
CO 3	Analyse historical background of planning/ environmental acts and rules also about various institutional frameworks.	Understanding and Analysing		
CO 4	Orientation towards existing rules, acts and legislation in the field of planning/environment.	Understanding and Analysing		
CO 5	Apply legal concepts towards addressing urban development.	Analysing and applying		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100 Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will				

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carry 10 marks and can have sub questions.

Module	Contents	CO	Hours
I	Introduction to legislation & terminology Constitution of India Concept and contents, provisions, regarding property rights and features. Concept of Law: Source of law (i.e., custom. legislation and precedent), meaning of terms of law, legislation, ordinance, Bill, Act, Regulations and Bye-laws.; Significance of law and its relationship to Urban planning benefit of statutory provisions- eminent domain & police powers.	CO1, CO2	5
II	Evolution of planning legislation & Institutional framework A brief History of Planning Legislation in India and Abroad. The Planning Commission, National Development Council, NITI Ayog, TCPO and their role in framing development policies. Other institutional framework for urban and rural development planning at national level, state level, regional/ district level.	CO2, CO3	5
III	Various planning acts and legislation in India Town and Country Planning Act - Its implication on preparation and implementation of regional plans, development plans, town planning schemes, area plans etc., Town and country planning act, Kerala. Critical Appraisal of: Improvement Trust Act, Development Authorities Act, Land Acquisition Act 1986, Urban Land (Ceiling and Regulation) Act 1976, Slum Clearance/Slum Improvement Act, Rent Control Act, 73rd and 74th Constitutional Amendment Acts, Right to Fair Compensation and Transparency in Land Acquisition and Rehabilitation and Resettlement Act, 2013 . Legislations for public-private partnerships in urban development and infrastructure projects. Regulations relating to development of integrated townships, Special economic zones, export processing zones, and IT Parks – tools to control: zoning, sub-division building regulations and bye-laws.	CO4, CO5	8
IV	Concept of environmental planning	CO2, CO3	6

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	Concepts of Environmental Planning, History of Environmental Planning, Global Concerns, Development of habitat patterns, settlement structure and form in response to environmental challenges. Components of environment – Classification of environmental resources - Purpose and objectives in environmental protection, planning and management – Consequence of development over urban and rural settlements		
V	Environmental Legislation Introduction to Law (Indian Constitution, Evolution of Environmental Laws in India) EP Act 1986; Air (Prevention and Control of pollution) Act Water (Prevention and Control of pollution) Act Mines and Mineral Act; Factories Act Pesticides Act. Critical Appraisal of: legislations on Coastal Zone regulations, Indian Forest act, Wildlife Act, Hill area regulations; Ancient Monuments and Archaeological Sites and Remains Act, Hazardous Waste Management and Handling Rules / Biomedical Rules / Solid Waste Management Rules; Environment Tribunal Act; Climate change Protocols and Conventions MOEF Guidelines and Notifications Appellate Authority Act and Other related Notifications.	CO4, CO5	8

References:

1. Divan, S., & Rosencranz, A. (2022). *Environmental law and policy in India: Cases and materials* (3rd ed.). Oxford University Press.
2. Doabia, T. S. (2023). *Environmental and pollution laws in India* (Vols. 1–3). LexisNexis.
3. Shah, U. C. (2023). *Planning legislation covering urban planning and environmental laws in India*.
4. Pai Panandiker, V. A., Mishra, S. N., Mishra, S., & Pai, C. (2000). *Governmental systems and development; Decentralised planning and Panchayati Raj institutions*. Mittal Publications.
5. Government of India. (2015). *Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines* (Vols. I & II). Town and Country Planning Organization, Ministry of Urban Development.
6. Salzman, J. E. (2019). *Environmental law and policy* (Concepts and Insights series). Foundation Press.
7. Randolph, J. (2012). *Environmental land use planning and management* (2nd ed.). Island Press.
8. Government of India. (1992). *The Constitution (73rd and 74th Amendment) Act*.
9. Nath, V. (n.d.). *Administration and development planning in India*.
10. Tyagi, R. (n.d.). *Public administration: Principles and practices*. Atma Ram & Sons.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252TPL003	Urban Economy and Employment	1-1-0	2	2025
Course Objectives	- The aim is to provide students with a comprehensive understanding of how urban economic principles intersect with urban planning, emphasizing the dynamics of labor markets, spatial development, and policy-making in cities. - This course equips students with analytical and practical skills necessary to address contemporary economic and employment challenges in urban settings, fostering inclusive, sustainable, and resilient urban economies.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Comprehend the definition, nature, scope, and key theories of urban economics. Understand basic urban economic terms and concepts.	Understanding		
CO 2	Understand the basic economic factors that affect the growth of urban areas, including the roles of policies and informal sectors.	Understanding		
CO 3	Assess the key requirements of land economics in the context of urban planning and development.	Analyzing		
CO 4	Investigate how socio-economic factors shape employment patterns, guided by key economic development initiatives and policies.	Analyzing		
CO 5	Prepare the student to engage in execution of economic planning and development in developing nations, and critically evaluate the role of programs, budgets, and institutions in poverty reduction and inclusive growth in India.	Applying		
Mark Distribution: Continuous internal evaluation - 40				

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End semester examination	- 60 (Duration - 2.5 hours)
Total Marks	- 100
Continuous Internal Evaluation pattern:	
Assignment/ Quiz/ Course project	- 25 marks
Internal test	- 15 marks
End Semester Examination pattern:	
There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.	

COURSE PLAN			
Module	Contents	CO	Hours
I	Introduction to Urban Economics Definition and Nature of urban economics, Scope and theories, Concepts of urban economics and urban economic growth development, Terms used in economics related to urban and regional Planning (URP). Central Problems of Economics, scarcity, opportunity cost, supply and demand, and resource allocation.	CO1	8
II	Fundamentals of Urban Economy in Planning Basic concepts of macro-and microeconomics- GDP, GNP, per capita income, inclusive development, law of demand and supply, types of demand, production economics, Urban and regional economic growth trends and patterns Economies of scale; urban and regional growth- Introduction to economics of urban areas. Economic policies and spatial planning in India, economic transformation in Indian cities and regions.	CO2	8
III	Land Economy Land use, Land Allocation, Market Value, Land demand, Land supply, Land use efficiency. Land use and land values: market mechanism and land use pattern, Theory of rent, welfare aspect of land rent, Location theories in intra-regional and inter regional contexts. Land speculation, taxation, and regulation, Role of land economics in urban planning and policy-making.	CO3	12

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IV	Socio Economic dynamism and employment Relationship between economic structure and social transformation, Migration patterns and impacts on city economics, formal and informal employment in cities, Sectoral employment trends, Occupational shifts and changing skill demands, Urban policies and Employment generation - Smart Cities Mission, AMRUT, Role of Special Economic Zones (SEZs) Digital livelihoods and Urban e- commerce.	CO4 CO5	10
V	Economic Planning and Development Economic Planning and programs to reduce the poverty, Budgets in India, engaging stakeholders and protecting rights, Role of NGO's, Banks and other institutions in economic planning in India	CO4 CO5	7

References:

1. O'Sullivan, A. (2018). *Urban economics*. McGraw-Hill Education. Misra, S. K., & Puri, V. K. (n.d.). *Indian economy*. Himalaya Publishing House.
2. Kapila, U. (n.d.). *Indian economy: Performance and policies*. Academic Foundation.
3. Chapin, F. S. (1965). *Urban land use planning*. Higg & Brothers.
4. Alonso, W. (1964). *Location and land use: Toward a general theory of land rent*. Harvard University Press.
5. McCann, P. (n.d.). *Urban and regional economics*. Oxford University Press.
6. Mankiw, N. G. (n.d.). *Principles of economics*. Cengage Learning.

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COURSE CODE	COURSE NAME	L-T-S	CREDIT S	YEAR OF INTRODUCTION
252TPL004	Urban Utilities and Infrastructure Planning	2-1-0	3	2025
Course Objectives	This course provides students with a thorough understanding of the various components, processes, and funding mechanisms essential to infrastructure development, highlighting both qualitative and quantitative impacts as well as their wider social and economic implications. Through a multidisciplinary approach, students cultivate critical analytical skills for assessing infrastructure projects, optimizing resource allocation, and navigating complex multi-service provider environments across different sectors. By merging theoretical foundations with practical methodologies, the course nurtures expertise in planning, execution, and policy considerations, equipping students to tackle evolving challenges in infrastructure systems with strategic insight and professional competence.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Comprehend the concept and typology of urban utilities infrastructure.	Understanding		
CO 2	Understand the various agencies and financial institutions involved in the infrastructure planning process.	Understanding		
CO 3	Assessing the actual requirements of different urban utilities and infrastructure facilities.	Analyzing		
CO 4	Investigate the different financial mechanisms and methods involved in project planning.	Analyzing		
CO 5	To engage in the planning and execution of diverse infrastructure projects in developing nations by analyzing and examining case studies related to various infrastructure initiatives in these regions.	Applying		
Mark Distribution:				

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Continuous internal evaluation - 40
End semester examination - 60 (Duration - 2.5 hours)
Total Marks - 100

Continuous Internal Evaluation pattern:

Assignment/ Quiz/ Course project - 25 marks
Internal test - 15 marks

End Semester Examination pattern:

There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

COURSE PLAN			
Module	Contents	CO	Hours
I	Concept and typology of infrastructure Infrastructure Planning: Basic Definitions, Concepts, Significance. The Role of Infrastructure Planning in National Development. Types of Infrastructure :Physical, Social, Economic Infrastructure, Utilities and Services, Urban Infrastructure, Smart City Infrastructure. Understand the roles of different actors in urban utility and infrastructure projects, and the government and NGOs that deliver urban utility and infrastructure services in India Requirement of infrastructure facilities, including components and qualitative and quantitative techniques for assessing requirements. - norms and standards URDPFI and CPHEEO	CO1	9
II	Planning and Management of Water, Sanitation, and Stormwater-Physical Infrastructure Water – sources of water, treatment and storage, transportation and distribution, quality, networks, distribution losses, water harvesting, recycling and reuse, norms and standards of provision. Case study Sanitation – generation points, collection, treatment, disposal, norms and standards, grey water disposal, DEWATS, Case study Storm water – rainfall data interpretation, points of water stagnation, system of natural drains, surface topography and soil characteristics, ground water replenishment, storm water collection and disposal,	CO2, CO3	11

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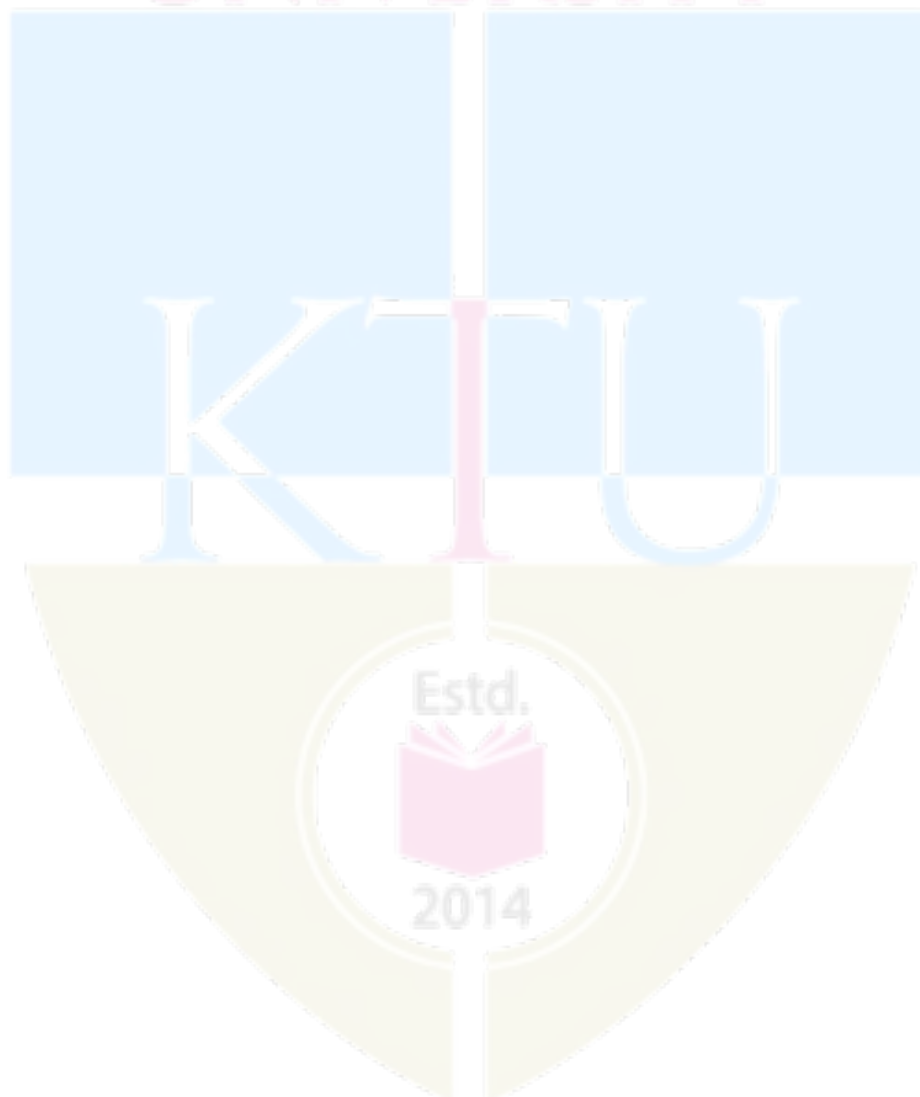
	norms and standards, case study Note ; Case study examines the critical management challenges, institutional frameworks, and strategic planning processes associated with each urban utility and infrastructure projects		
III	Planning and Management of Municipal Wastes, Power, and Fire Municipal and other wastes – generation, typology, quantity, collection, storage, transportation, treatment, disposal, recycling and reuse, wealth from waste, norms and standards, case study Power – Sources of power procurement, distribution networks, demand assessment, norms and standards, Case study Fire – vulnerable locations, methods of firefighting, norms and standards, case study related to specific infrastructure Note; Case study examines the critical management challenges, institutional frameworks, and strategic planning processes associated with each urban utility and infrastructure projects	CO2 CO3	12
IV	Social and economic Infrastructure Education, Health, Civic Types, hierarchical distribution of facilities, Access to facilities, provision and location criteria, Norms and standards, Infrastructure Management: Planning Issues -norms and standards – case study from developed and developing nations	CO2 CO3	7
V	Financial mechanism for urban utility and infrastructure planning projects Understand the terms: Equity, Efficiency, Quality of Service, Paying Capacity, and Pricing of Infrastructure Services. Ownership and Control: Public, Private, Special Purpose Vehicle (SPV), and various Public-Private Partnership (PPP) models and their collaboration in infrastructure Provision, Emerging opportunities in infrastructure development: case studies from Asian cities showcasing innovative solutions and equitable economic growth.	CO4 CO5	9

References:

1. Amekudzi, A. A., & McNeil, S. (2008). *Infrastructure reporting and asset management: Best practices and opportunities*. American Society of Civil Engineers.

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2. India Infrastructure Reports. (n.d.).
3. Stein, J. M. (1988). *Public infrastructure planning and management*. Sage Publications.
4. Schübeler, P. (1996). *Participation and partnership in urban infrastructure management*. World Bank Publications.
5. Government of India. (2015). *Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines (Vol. I)*. Ministry of Urban Development, Town and Country Planning Organization.
6. CPHEEO. (n.d.). *Manuals on water supply, sewerage, and solid waste management*. Central Public Health and Environmental Engineering Organisation, Ministry of Housing and Urban Affairs, Government of India.



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252LPL001	GIS Applications II	0-2-0	2	2025
Course Objectives	This course aims to enable students to learn and apply GIS during their academics and equip them to become proficient in the professional practice of applying GIS.			
Course Outcomes: After the completion of the course, the students shall be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Design and manage spatial databases to support complex planning processes and integrate multi-source spatial and non-spatial data.	Applying and Creating		
CO 2	Perform advanced spatial analysis and modelling, including network analysis, spatial statistics, and geostatistics, to understand urban patterns and dynamics.	Analyzing and Evaluating		
CO 3	Apply GIS-based Multi-Criteria Decision Analysis (MCDA) and site suitability techniques to support evidence-based decision-making in planning.	Evaluating and Creating		
CO 4	Execute a GIS-based planning project, demonstrating proficiency in spatial problem-solving, map-making, and communicating planning solutions using geospatial tools.	Evaluating and Creating		
Mark Distribution: Continuous internal evaluation - 40 marks				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 20 marks Internal test - 20 marks				

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COURSE PLAN			
Module	Contents	CO	Hours
I	Urban Growth Modelling & Scenario Planning - Land-use change detection (remote sensing: NDVI, urban sprawl). - Predictive modelling (Markov chains, cellular automata). - Climate resilience (flood risk, heat islands). - Analyse urban expansion using Landsat imagery (1990–2020). - Simulate future growth scenarios with plugins.	CO1, CO4	
II	Participatory GIS & Decision Support - Public participation GIS (PPGIS): Crowdsourcing data (e.g., KoboToolbox). 3D visualisation (SketchUp integration, TINs for terrain). - Dashboard creation (Power BI/ArcGIS Urban for planners). - Class project: GIS model of a proposed development in line with the studio, including some additional analysis.	CO1, CO2, CO3	

References:

1. Chang, K. T. (2019). *Introduction to geographic information systems* (9th ed.). McGraw-Hill Education.
2. Jensen, J. R. (2016). *Remote sensing of the environment: An earth resource perspective* (3rd ed.). Pearson.
3. Lillesand, T. M., Kiefer, R. W., & Chipman, J. W. (2020). *Remote sensing and image interpretation* (8th ed.). Wiley.
4. Lo, C. P., & Yeung, A. K. W. (2007). *Concepts and techniques of geographic information systems* (2nd ed.). Pearson.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252SPL001	Planning Studio II	0-0-10	5	2025
Course Objectives	• To understand the statutory framework and planning process governing the preparation of Development Plans. • To analyse the existing physical, socio-economic, environmental, and institutional structure of the study area to identify key developmental issues, potentials and constraints. • To formulate spatial strategies and landuse proposals for the study region, with due consideration of the intricacies inherent in the planning process • To develop zoning regulations, development control rules, and implementation frameworks in line with legal requirements.			
Course Outcomes: After the completion of the course, the students shall be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Critically analyse crucial factors, and statutory frameworks involved in development plan preparation	Understanding and analyzing		
CO 2	Apply planning techniques and prepare plans at the city/peri-urban level.	Applying		
CO 3	Assessing and integrating the various possible solutions to reach the best possible solution for a complex problem.	Applying		
Mark Distribution: Continuous internal evaluation -160 (+40 marks for GIS Lab) Final Jury Evaluation (Viva) - 100 Total Marks - 260 (300 marks including GIS Lab)				

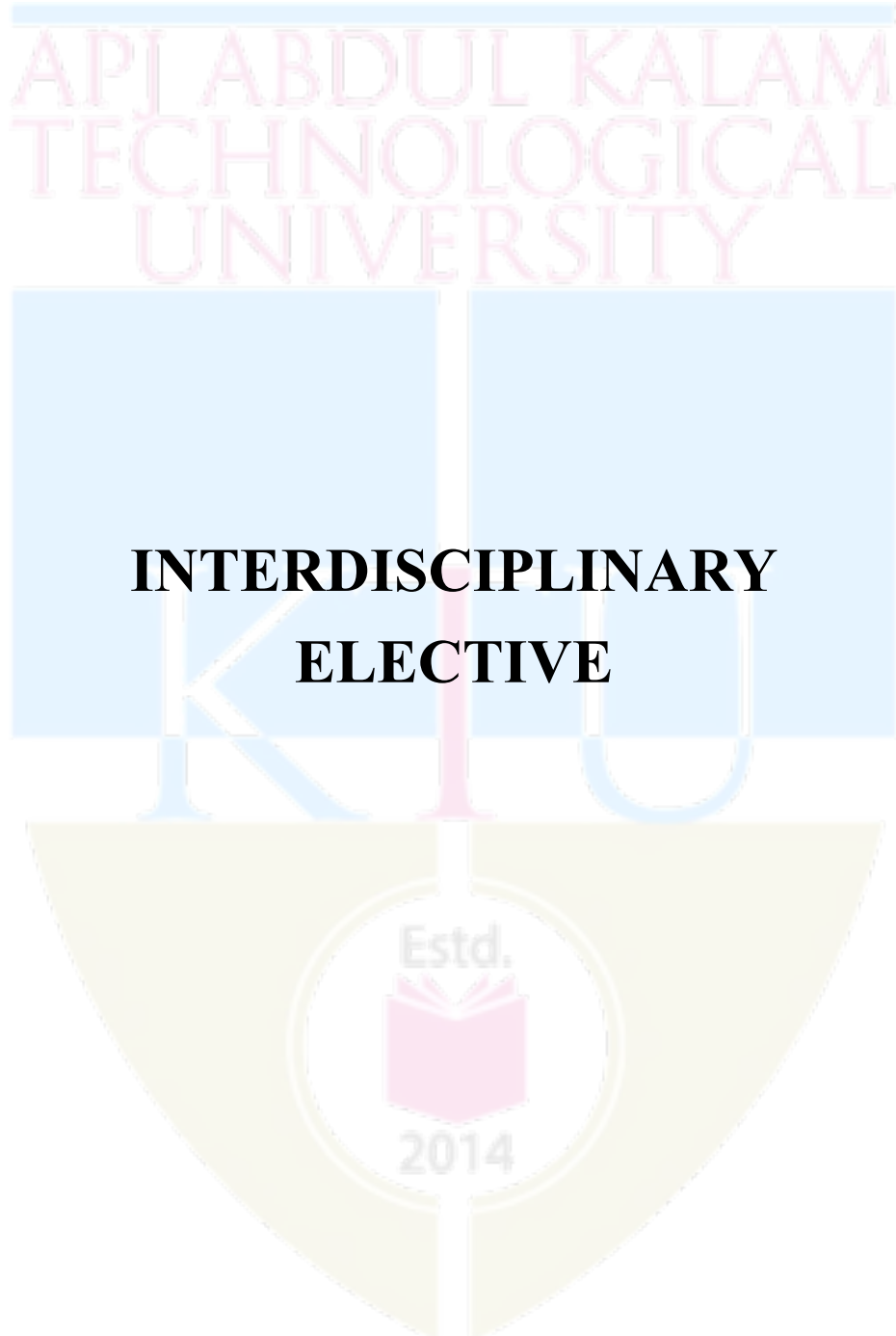
COURSE PLAN			
Module	Contents	CO	Hours
I	Development Planning Exercise Scale of the study area- One or more local-bodies within	CO1 CO2 CO3	160

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	urban/peri-urban areas The basic aim of this project is to develop a comprehensive development plan for one or more local bodies having urban/peri-urban character. The development planning process should consider the local and regional context, in alignment with physical, socio-economic, environmental, institutional and legal framework and future growth trajectories. This process should also facilitate a multidisciplinary approach to addressing the numerous factors that must be considered in development planning to understand the city through various rational thinking methods. The project exposes the student to data collection, analytical research methodologies, field work, interaction with relevant stakeholders, teamwork, critical analysis, presentation of the study verbally and illustratively, and report preparation at a city/peri-urban level. Minor project should address the sector focused detailed plans and proposals. (Tourism/Industrial/or any other relevant sectors))		
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References:

1. Urban and Regional Development Plans Formulation & Implementation Guidelines (2014) Ministry of Urban Affairs & Employment, Govt. of India, New Delhi
2. Ewing, R., & Park, K. (Eds.). (2020). Basic Quantitative Research Methods for Urban Planners (1st ed.). Routledge. <https://doi.org/10.4324/9780429325021>
3. Klosterman, Richard E., Brooks, Kerry, Drucker, Joshua, Feser, Edward และ Renski, Henry (2561). Planning support methods: urban and regional analysis and projection. Rowman & Littlefield.



**INTERDISCIPLINARY
ELECTIVE**

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EPL001	Barrier free Built Environment	2-1-0	3	2025
Course Objectives	The course aims to develop an understanding of barrier free built environment which aims to include all sections of the society including Persons with disabilities and the elderly. It develops a sensitivity to view the practice of building construction critically to analyse the subtle way in which it is marginalising the underprivileged sections. It also aims to familiarise the students with relevant acts, codes and standards mandatory for ensuring inclusion and livability.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	To develop a conceptual understanding of disability, accessibility and inclusion	Understanding		
CO 2	To introduce the legal and regulatory framework relating to barrier free built environment	Understanding		
CO 3	To familiarise students with the difficulties faced by Persons with disabilities	Analyzing		
CO 4	To enable students to create a barrier free built environment through architecture, design and planning.	Creating		
CO 5	To equip students to conduct access audit and design for retrofitting	Creating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions. (Students should be permitted to refer all relevant codes while writing the examinations)				

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Module	Contents	CO	Hours
I	Understanding the concepts of accessibility, disability and inclusion Definition of universal accessibility Importance of inclusivity in built environment Historical context and evolution of accessibility laws, UNCRPD Rights of Persons with Disabilities legislations, guidelines and standards in India Universal Design Principles Barriers and Accessibility Challenges Types of disabilities and understanding the needs of people with various disabilities (mobility, visual, auditory, cognitive) Accessibility in public and private spaces Case studies of inaccessible spaces	CO1, CO2, CO3	11
II	Accessibility codes and standards for External Environment National Building Code (NBC), Harmonised Guidelines and other regulations Accessible Pathways and Circulation Designing ramps and stairways Clear and safe circulation routes Rest areas, waiting zones and bus stops Accessible parking Importance of Accessibility in employment, education, leisure and other activities Designing with TGSIs	CO2, CO3, CO4	11
III	Accessibility codes and standards for internal environment Accessible Entrances and Exits Automatic doors, thresholds and doorways Door hardware and controls Emergency exits and evacuation routes for individuals with disabilities Accessible toilet design and layout, fixtures, signage, and grab bars, shower and bathing accessibility Accessible public facilities- Auditorium, commercial areas, offices Detailing for accessibility and safety- handrails, edge strips etc.	CO3, CO4	9
IV	Accessibility of Transportation system and the digital environment Accessible transport systems, and transit stations Smart technologies in accessible design (assistive listening devices, virtual tours, etc.) Interactive and adaptive systems in buildings	CO2, CO4	9

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	Accessible digital spaces (websites, apps, kiosks) Accessible Signage and Wayfinding Codes and standards for accessible websites and digital spaces.		
V	Best Practices of Accessibility Case studies of accessible facilities Design of special buildings- care homes, special schools, assistive living. Interviews with persons with disabilities and elderly Accessibility Audit Best practices in other countries	CO3, CO5	8

References:

1. Bureau of Indian Standards. (2016). *National Building Code of India 2016* (Vols. 1–2). Bureau of Indian Standards.
2. Ministry of Housing and Urban Affairs. (2021). *Harmonised guidelines and standards for universal accessibility in India 2021*. Government of India.
3. Telecommunication Engineering Centre. (2018). *Accessibility standards for ICT products and services – Part I & II*. Department of Telecommunications, Government of India.
4. Raffaelli, B. (2020). *The fast guide to accessibility design*. BIS Publishers.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EPL002	Circular Economy	2-1-0	3	2025
Course Objectives	Understand Core Concepts, Analyze Environmental and Economic Impacts, Apply Circular Strategies, Integrate Circular Thinking into Planning and Design, Evaluate Policy and Regulatory Frameworks, Use Analytical Tools, Explore Real-World Applications, Develop Critical and Systems Thinking.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand circular economy fundamentals.	Understanding		
CO 2	Analyze strategies in various sectors explaining their relevance to urban development.	Understanding		
CO 3	Apply the principles of Circular Economy to real-world urban planning projects. Apply tools like MFA and LCA.	Applying		
CO 4	Evaluate the success of existing Zero-Waste Cities comparing and contrasting case studies. Propose circular economy solutions.	Evaluating, creating		
CO 5	Critically assess the challenges and opportunities of implementing Circular Economy practices in diverse socio economic and cultural contexts. Evaluate policies and governance frameworks.	Evaluating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2				

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questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

COURSE PLAN			
Module	Contents	CO	Hours
I	Fundamentals of circular economy Linear economy, economic and ecological disadvantages of linear economy. Replacing linear economy with circular economy, Development of concept of circular economy and its relevance to cities, Circular Vs Linear economy, Circular economy in urban planning, CE and Sustainable Development Goals (SDGs), acknowledging the challenges and barriers to its implementation.	CO1	4
II	Circular urban design and planning principles. Material recovery, Waste Reduction, reducing negative externalities, Explaining Butterfly diagram, Concept of Loops, Urban land use and zoning for circularity, Compact city, mixed-use planning, transit-oriented development, Urban commons and shared infrastructure, Green-blue infrastructure integration	CO2	4
III	Circular resource flows in cities Zero waste: Principles of Zero-Waste Urbanism, Waste Management in context of Circular Economy, Water reuse and circular sanitation, Recycling, Upcycling & Waste-to-Energy, Circular construction: use of recycled materials, modular design, Material Flow Analysis (MFA) and Life Cycle Assessment (LCA), Renewable energy and energy-positive buildings.	CO3	8
IV	Applications in urban context /case studies & city transitions. Adaptive Reuse and Retrofitting in Urban Development, Cradle-to-Cradle Buildings, Extended Producer Responsibility (EPR) & Reverse Logistics. Circular food systems and urban agriculture, Mobility: shared and electric transport systems, Urban manufacturing and repair economy, Case Studies-Global cities: San Francisco, Amsterdam, Copenhagen, Singapore, etc. Indian examples: Panaji, Indore, Pune, and Smart Cities Mission.	CO4	9
V	Policies, governance, and future trends in circular urbanism	CO5	5

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	CE policies at global (EU, UN-Habitat), national (India's Resource Efficiency), and city levels, Financing Mechanisms for Circular Urban Infrastructure, Smart Technologies (IoT, AI) for Resource Efficiency, Challenges & Opportunities in Mainstreaming Circular Urbanism, Future Directions: Doughnut Economics, Bio-Based Urban Systems		
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References:

1. Bhalla, S. G. (n.d.). *Circular economy: (Re) emerging movement*. Invincible Publishers.
2. Ellen MacArthur Foundation. (n.d.). *Circular economy in cities*. <https://ellenmacarthurfoundation.org>
3. European Commission. (2020). *Circular cities and regions initiative (CCRI)*. <https://ec.europa.eu>
4. Franco-García, M.-L., Carpio-Aguilar, J. C., & Bressers, H. (2019). *Towards zero waste: Circular economy boost, waste to resources*. Springer International Publishing.
5. Ghosh, S. K. (Ed.). (2020). *Circular economy: Global perspective*. Springer.
6. Girardet, H. (2015). *Creating regenerative cities*. Routledge.
7. Lacy, P., Long, J., & Spindler, W. (2020). *The circular economy handbook: Realizing the circular advantage*. Palgrave Macmillan.
8. Lacy, P., & Rutqvist, J. (2015). *Waste to wealth: The circular economy advantage*. Palgrave Macmillan.
9. Liu, L., & Ramakrishna, S. (2021). *An introduction to circular economy*. Springer Singapore.
10. Ministry of Housing and Urban Affairs, Government of India. (n.d.). *Reports and publications on urban development and circular economy*. <https://mohua.gov.in>
11. Stahel, W. R. (2019). *The circular economy: A user's guide* (1st ed.). Routledge.
12. Tonelli, M., & Cristoni, N. (2018). *Strategic management and the circular economy*. Routledge.
13. UN-Habitat. (2019). *Waste Wise Cities toolkit: Step-by-step guide for municipal solid waste management in cities*. United Nations Human Settlements Programme (UN-Habitat). <https://unhabitat.org>

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EPL003	Urban Analytics	2-1-0	3	2025
Course Objectives	The goal of this course is to equip urban planning students with the knowledge and practical skills needed to understand, analyze, and apply big data and data science tools in the context of urban development and smart cities. It aims to bridge traditional planning approaches with modern data-driven methods, enabling students to make informed, evidence-based decisions using spatial, statistical, and real-time data. The course fosters the ability to interpret complex urban systems through visualization, mapping, and analytics, preparing students to contribute effectively to contemporary urban governance and policy-making.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	To develop a conceptual understanding of big data, data science, and the smart cities framework in urban planning.	Understanding		
CO 2	To familiarize students with types of urban datasets and ethical considerations in data use.	Understanding		
CO 3	To enable students to use spatial tools to represent and analyze urban issues.	Analyzing		
CO 4	To provide hands-on experience in managing and analyzing urban datasets.	Applying		
CO 5	To synthesize course learning into a project that addresses a real-world urban issue using data.	Creating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in				

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this part will carry 10 marks and can have sub questions.

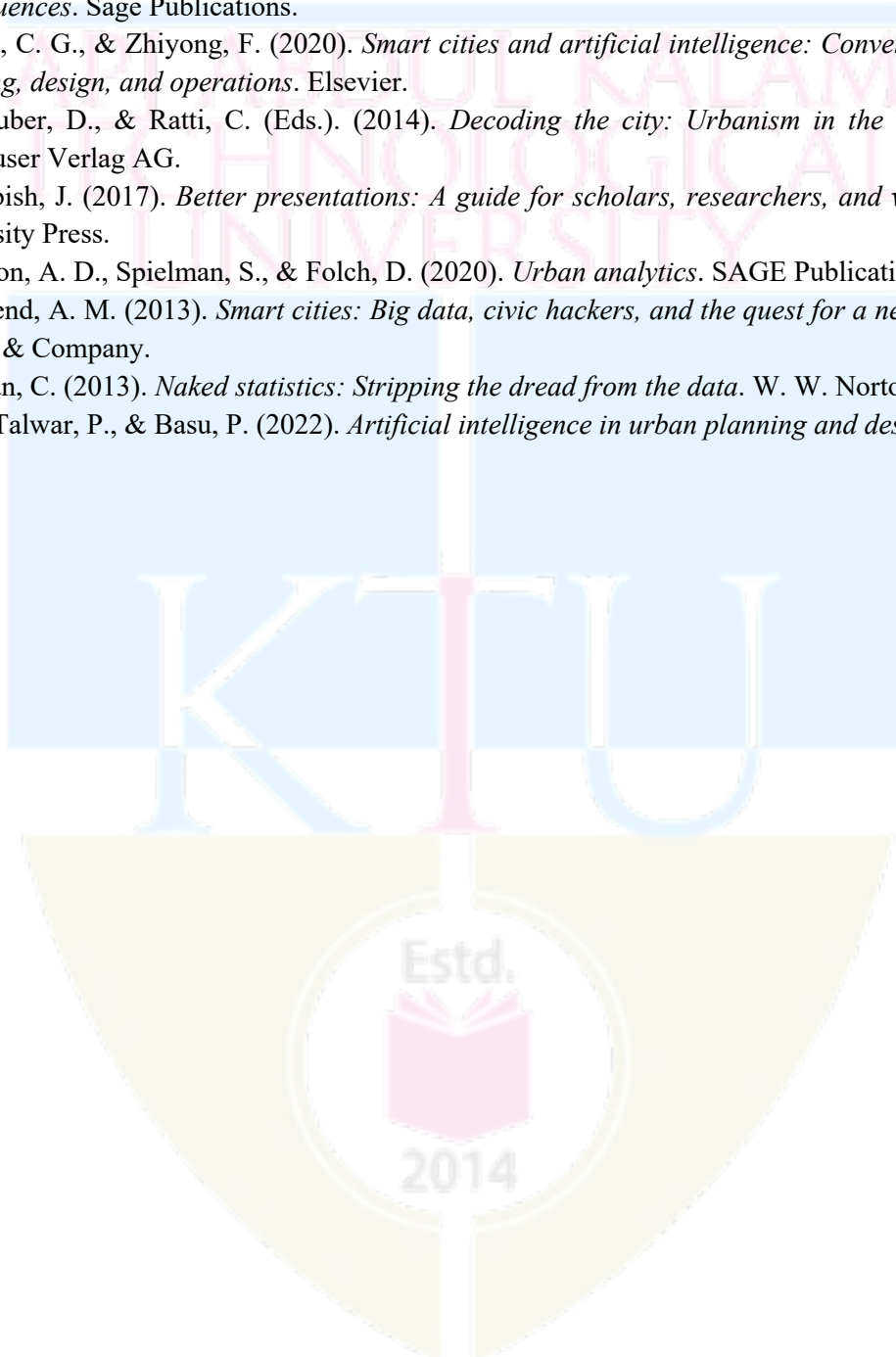
Module	Contents	CO	Hours
I	Foundations of Urban Data Science and Smart Cities Big data concepts and characteristics, Emergence of data science in urban planning, Smart cities and digital governance, Importance of data-driven decision-making, Urban systems and ICT applications. NIUA, smartcity dashboard; Open City urban Data Portal	CO1	8
II	Urban Data Sources, Ethics and Policy Census and economic datasets, IoT, Open data platforms and civic technologies, Real-time data sources and sensors, Data privacy, security, and ethics, Digital inclusion and legal regulatory policies and frameworks	CO2	10
III	Spatial Mapping, Visualization & Digital Twin GIS fundamentals (QGIS, ArcGIS), Map elements: scale, legend, projection, Online mapping platforms (OpenStreetMap, Mapbox, kepler.gl), Story maps and thematic mapping, Visualizing urban inequalities and trends.	CO3, CO4	10
IV	Data Handling and Analytics Data cleaning and pre-processing, SQL for data querying, Programming basics in R or Python for urban data, Statistical analysis methods Visualization tools: Tableau, Power BI, ggplot2, Google Earth Engine. AI, ML and Predictive Urban Analytics Basics of ML, Predictive analytics, Clustering, Forecasting urban growth, Mobility patterns, Satellite image classification (AI for remote sensing).	CO2, CO4	10
V	Data-Driven Urban Planning Solutions Defining research questions and project goals, Data sourcing and integration, Analytical methods and visualization. Report writing and presentation, Applied urban data science for planning solutions.	CO4, CO5	10

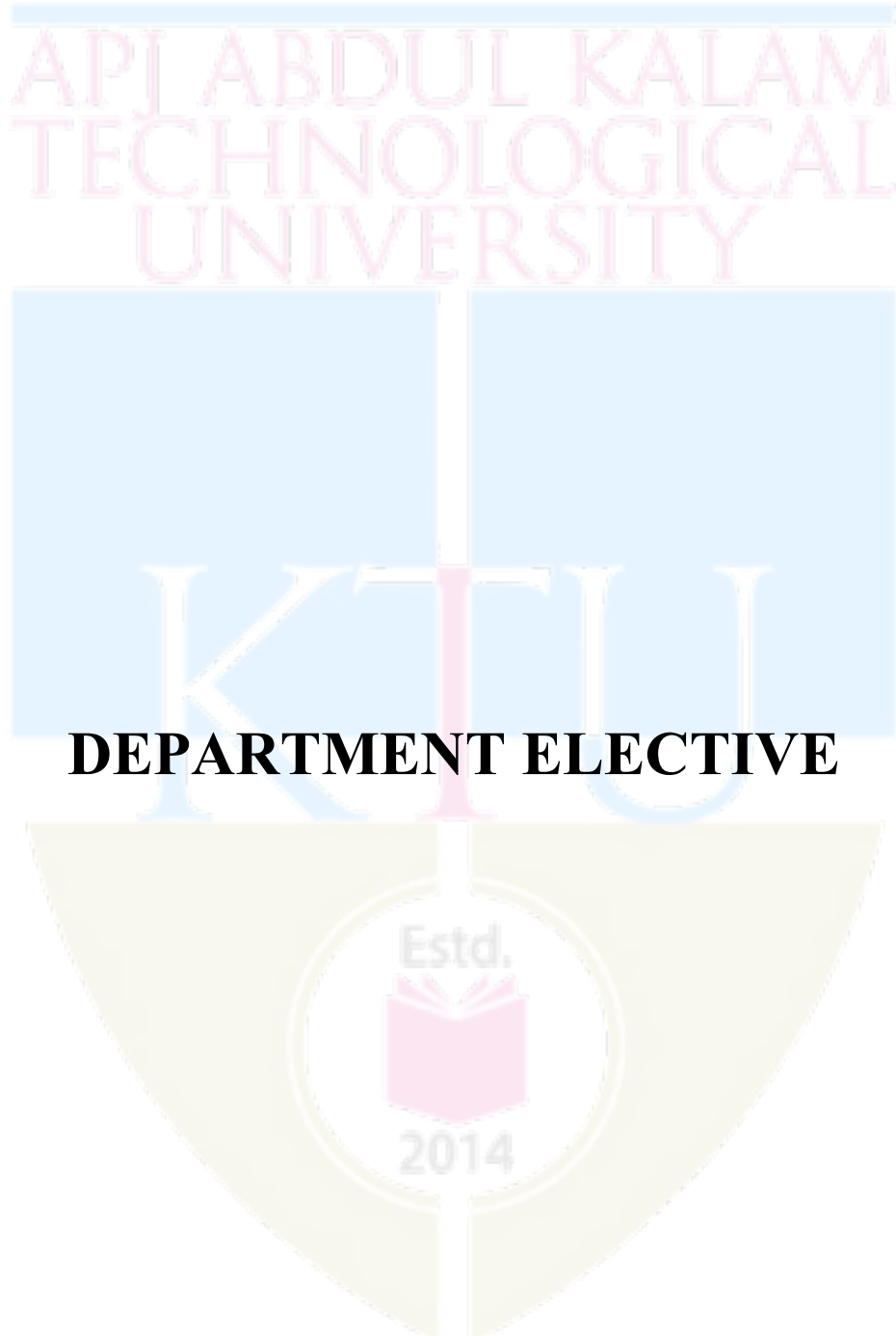
References:

1. Batty, M. (2013). *The new science of cities*. MIT Press.
2. Carta, S. (2019). *Big data, code and the discrete city: Shaping public realms*. Routledge.

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3. Desouza, K., & Smith, K. (2016). *Big data and planning* (PAS Report No. 585). American Planning Association.
4. Grimaldi, D., & Carrasco-Farré, C. (2021). *Implementing data-driven strategies in smart cities: A roadmap for urban transformation*. Elsevier.
5. Kitchin, R. (2014). *The data revolution: Big data, open data, data infrastructures and their consequences*. Sage Publications.
6. Kirwan, C. G., & Zhiyong, F. (2020). *Smart cities and artificial intelligence: Convergent systems for planning, design, and operations*. Elsevier.
7. Offenhuber, D., & Ratti, C. (Eds.). (2014). *Decoding the city: Urbanism in the age of big data*. Birkhäuser Verlag AG.
8. Schwabish, J. (2017). *Better presentations: A guide for scholars, researchers, and wonks*. Columbia University Press.
9. Singleton, A. D., Spielman, S., & Folch, D. (2020). *Urban analytics*. SAGE Publications.
10. Townsend, A. M. (2013). *Smart cities: Big data, civic hackers, and the quest for a new utopia*. W. W. Norton & Company.
11. Wheelan, C. (2013). *Naked statistics: Stripping the dread from the data*. W. W. Norton & Company.
12. As, I., Talwar, P., & Basu, P. (2022). *Artificial intelligence in urban planning and design*





DEPARTMENT ELECTIVE

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EAR001	Urban Conservation and regeneration	1-1-0	2	2025
Course Objectives	This course aims to provide a foundational understanding of heritage and conservation principles, including the ethical and cultural dimensions of preserving built environments. It enables students to analyze urban heritage, historic settlements, and cultural landscapes, and equips them with skills in documentation, assessment, and planning of conservation interventions. The course also emphasizes the evaluation of legal frameworks, policies, and institutional roles, while encouraging critical reflection through case studies to develop context-sensitive conservation and regeneration strategies.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand the core concepts, ethics, and evolution of urban conservation and heritage management.	Remembering, Understanding		
CO 2	Analyze typologies and character of historic settlements, urban heritage, and cultural landscapes.	Analyzing		
CO 3	Apply techniques of heritage documentation, assessment, and planning in urban conservation practice.	Applying		
CO 4	Evaluate tools, legal frameworks, and institutional roles in conservation and regeneration processes.	Evaluating		
CO 5	Critically assess case studies and propose context-sensitive conservation or regeneration strategies.	Analyzing and Evaluating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern:				

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There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

Module	Contents	CO	Hours
I	Fundamentals of Conservation Definitions: Tangible and intangible heritage, built and cultural heritage, conservation vs. preservation: ethics, principles, degrees of intervention. Distinction between architectural and urban conservation. Concepts of authenticity, integrity, and cultural landscapes. International and Indian conservation paradigms.	CO1, CO2	6
II	Urban Heritage and Historic Landscapes Components of urban heritage: morphology, public spaces, uses, cultural practices and traditions Historic cities and precincts: Typologies Concept of historic precincts and cities, Cultural Landscape, Historic Urban Landscape (HUL), charters pertaining to them and case examples. World Heritage Cities : Case examples and criteria Historic cities : Indian examples, Case studies to identify components and Issues in Historic cities/cores.	CO1, CO2, CO5	6
III	Conservation Process and Practices Urban conservation process from archival research to management plans. Tools and Techniques: Inventories, photogrammetry, mapping, visual surveys,	CO2, CO3, CO5	6

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	Heritage Impact Assessments (HIAs) Case examples of interventions in Historic cores : Adaptive reuse, restoration, rehabilitation, infill in historic areas. Role of public participation, and heritage-led regeneration.		
IV	Tools, Legislations, and Institutional Frameworks Legal instruments: AMASR Act, CRZ, DCRs, Wetlands Rules, Masterplans: Special zones National, state institutions and NGOs: ASI, INTACH, UAC, MoHUA Conservation schemes, funding models, and urban policy integration - AMRUT, HRIDAY, JNNURM, CSR in Heritage conservation Role of ULBs, state departments, community groups Heritage and Tourism : Prospects and problems	CO1, CO4, CO5	7
V	Urban Regeneration and Case Studies Concepts: Redevelopment, renewal, rejuvenation, reconstruction Triggers for regeneration: degradation, obsolescence, socio-economic shifts Planning and implementation strategies: phasing, financing, legal tools Comparative regeneration models: East and West Case studies: Global and National.	CO3, CO4, CO5	7

Reference Books

1. Feilden, B. M. (2003). *Conservation of historic buildings* (3rd ed.). Routledge.
2. Glendinning, M. (2013). *The conservation movement: A history of architectural preservation*. Routledge.
3. Menon, A. G. K. (Ed.). (n.d.). *Conservation of immovable sites*. INTACH Publication.
4. Hollis, M. (2000). *Surveying buildings*. RICS Business Services Limited.
5. Silva, K. D., & Chapagain, N. K. (2013). *Asian heritage management: Contexts, concerns and prospects*. Routledge.
(Please confirm publisher and year; Routledge is common for this title.)
6. INTACH. (n.d.). *Heritage and conservation: An introductory handbook*. INTACH.
7. Banga, I. (2005). *The city in Indian history*. Manohar Publishers and Distributors.
8. ICOMOS, & UNESCO. (n.d.). *Charters and guidelines*.
(If available, specify edition or publication year.)

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9. Cody, J., & Siravo, F. (2019). *Historic cities: Issues in urban conservation*. Getty Conservation Institute.
10. Ministry of Housing and Urban Affairs, Government of India. (n.d.). *HRIDAY reflections: A monograph on the heritage city development and augmentation Yojana*.



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EAR002	Environmental Impact Assessment	1-1-0	2	2025
Course Objectives	This course aims to determine and forecast the impact of a project on the biological, geophysical, and chemical environment, as well as human health, and suggest suitable policies, initiatives, and operational practices to mitigate this impact.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Identify both favourable and unfavourable potential environmental effects of a given project. Analyse how the "short-term" and the “long-term" time frames are used to identify these effects.	Remember, Analyse and		
CO 2	Formulate strategies that, when implemented, will mitigate or offset the adverse environmental impacts of a project that causes minimal environmental degradation.	Create		
CO 3	To assess how well environmental protection regulations are being implemented and how beneficial they are.	Evaluate		
CO 4	To prepare EIA reports with a collaborative approach	Create		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2				

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questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

Module	Contents	CO	Hours
I	Fundamental Approach to Environmental Impact Assessment: Definition, principles, and objectives, Historical evolution Legislation and Authorities concerning EIA: National & International EIA Methodologies: checklists, matrices, and modelling	CO1	6
II	Prediction and Assessment of Impacts on Soil and Groundwater Environment, Surface Water Environment, & Air Environment. Case Studies and Cross-Media Effects	CO1 CO3	6
III	Prediction and Assessment of Impacts on Biological Environment, Noise on the Environment, Socio-Economic and Human Health Impacts, Community engagement.	CO3	6
IV	Environmental Risk Assessment (ERA) and Risk Management in EIA, Application for Remote Sensing and GIS for EIA, Quantitative and Qualitative Risk Methods	CO3	7
V	EIA Case Studies review and appraisal, EIA Classroom Project, Preparation of an EIA report, Presentation of EIA reports.	CO2 CO4	7

Reference Books

1. Lohani, B. N., et al. (1997). *Environmental impact assessment for developing countries in Asia*. Asian Development Bank.
2. Raman, N. S., et al. (2014). *Environmental impact assessment*. TechSar Pvt Ltd.
3. Glasson, J., Therivel, R., & Chadwick, A. (2012). *Introduction to environmental impact assessment*. Taylor & Francis Ltd.
4. Wathern, P. (2015). *Environmental impact assessment: Theory and practice*. Routledge.
5. Satsangi, A., & Sharma, A. (2015). *Environmental impact assessment (EIA) and disaster management*. Rajat Publications.
6. Srivastava, A. K. (2012). *Environmental impact assessment: Theory and practice*. APH Publishing Corporation.

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7. Gilpin, A. (1994). *Environmental impact assessment: Cutting edge for the 21st century*. Cambridge University Press.
8. Das, S. (2024). *Environmental planning and impact assessment: Strategies for sustainable management*. Global Net Publication.
9. The World Bank. (1991). *The environmental assessment sourcebook* (Vols. I–III). The World Bank.
10. Anjaneyulu, Y., & Manickam, V. (2020). *Environmental impact assessment methodologies* (3rd ed.). BS Publications.
11. Reddy, P. J. (2016). *Environmental impact assessment theory & practice*. BS Publications



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EAR003	Urban Sociology	1-1-0	2	2025
Course Objectives	This course introduces key concepts of Urban Sociology and gives special attention to the emergence and growth of Urban Sociology, social structure in Urban society and challenges in urban governance. This course will provide an exposure to key theoretical perspectives for understanding urban phenomena in historical and contemporary contexts. With case studies from India and other parts of the world, this course will help students understand and relate to the complexities of urban living. This course seeks to evolve critical thinking and policy perspectives on sustainable urban development.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	To understand and analyze fundamental urban sociological concepts, its emergence and growth and understand types of cities with special emphasis on Indian cities.	Understanding		
CO 2	To critically analyze and apply theories of Urban sociology and theories of Urbanisation.	Applying		
CO 3	To understand Urban Social Structure, inequalities and polarization, Social area analysis and factorial ecology.	Understanding		
CO 4	To apply best practices of Urban governance to address Urban social problems	Applying		
CO 5	To analyze and understand social psychology and social geography and to address gender and equity concerns in urban development.	Understanding & applying		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				

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End Semester Examination pattern:

There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

Module	Contents	CO	Hours
I	Introduction to Urban Sociology Urban Sociology: meaning, definition, nature and scope. Basic concepts: Urbanization, Urbanism, Urbanity, Rural-Urban Fringe, Conurbation, Suburbanisation, Peri urbanisation, Ghettoisation, Urban neighbourhoods. Brief history of Urban Sociology: Origin and development of urban societies. Types of cities: Preindustrial City, Industrial City and Post-Industrial city; Examples. Indian cities: demographic, economic and ecological characteristics, case examples.	CO1	6
II	Theories of Urban Sociology and Urbanization Classical theories of City: Theories of Max Weber; George Simmel. The Chicago School of Sociology and its Critics: Theories of Robert Park, Burgess & Louis Wirth. Urban Anomie Theory: Emile Durkheim, Robert K. Merton. The New Urban Sociology: Henri Lefebvre, David Harvey, Manuel Castells. Theories of Urbanization: Theory of spatial disparity, Migration theories, Bid Rent theory, Modern theory of urbanization. Application of theories in contemporary urban contexts.	CO2	7
III	Urban Social Structure and Inequalities Urban Social Structure: definition, types of urban social structure, case examples Community in urban setting: identity & difference, Urban Culture & Urban activities Inequalities and Polarisation: patterns and types of inequality; Group Polarisation, Apartheid, Ethnic segregation of space, Gentrification, Marginalisation, Gender Inequality.	CO3	6

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	Social Area Analysis and Factorial Ecology, Case example.		
IV	Urban Social Problems and Urban Governance Urban Social Problems: Crime and Juvenile delinquency, Fear of Crime, Slum & Health, Poverty & Unemployment, Immigration, Beggary, Problems of women and children, Urban Violence, Terrorism in India. Solutions to urban social problems. Best Practices. Urban Governance: Participatory & Inclusive governance, Challenges of urban governance in Indian context, Measures to strengthen local governance to realise sustainable urban development.	CO4	6
V	Social Psychology and Social Geography Social Psychology: definition, basic concepts. Nature and scope of social psychology in Indian context. Theories of social psychology & Examples of Social psychological experiments. Social geography: Models of social geography, European perspectives of social geography, Image of the city and Mental mapping, examples. Planners and urban designers' role in empowering the image of cities. Ecofeminism, Production of space geography of gender in city, Case examples.	CO5	7

Reference Books

1. Flanagan, William G. *Urban Sociology: Images and Structures*, 5th Edition, Rowman and Littlefield Publishers Inc., New York, 2010.
2. Sharma K.R., *Urban Sociology*, New Delhi: Atlantic Publishers, 1997.
3. Bhushan, Vidya & D.R. Sachdeva. *An Introduction to Sociology*, Kutab Mahal distributors, 2014.
4. Burgess, Ernest W., and Robert E.Park, *The City*, Chicago, IL: University of Chicago Press, 1984.
5. M.Francis Abraham. *Modern Sociological Theory- An Introduction*, Delhi Oxford University Press, 1995.
6. Richard T.Schaeffer, Robert P.Lamm, *Sociology*, International edition, McGraw-Hill Inc., 2001.
7. James W. Vander Zanden. *The Social Experience-An Introduction to Sociology*, International Edition, McGraw Hill Publishing Co.,1990.
8. Michael, S.M. *Culture and Urbanization*, Delhi: Inter-India Publications.,1989.
9. Murray Lee, Gabe Mythen, *The Routledge International Handbook on Fear of Crime*, Taylor & Francis Group, 2017.
10. Carter, H., *The Study of Urban Geography*, Edward A. Old London, 1972.
11. Sibnath Deb, Anjali Gireesan, Pooja Prabhavalkar, Shayana Deb, *Social Psychology- Theories and Applications*, Routledge, New York, 2023.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
254MPL001	Elective I - MOOC	0-0-0-0	2	2025

Massive Open Online Course (MOOC) introduced and made mandatory for Master of Planning degree course is meant to provide opportunities for learners to get wider exposure, through flexible learning. The courses offered in varied topics are handled by top resource people, which are easily accessible and affordable, transcending geographical barriers.

Instructions for MOOC course

1. Students should choose one MOOC Certificate course from AICTE / SWAYAM or NPTEL. The MOOC can either be from the approved list given by the university or by the concerned college.
2. The MOOC should have a **minimum duration of 8 weeks**, and the content of the syllabus should be enough for **at least 40 hours of teaching**.
3. The course should include a **proctored/ offline examination** at the end of the course.
4. For securing the credits, the student is required to **complete the assessment** of the course and to provide the **certificate** of the course.
5. The fees (if any) for the registration and/or assessment of the MOOC course must be borne by the student only.
6. Even if the student completes more than one MOOC course, then **2 credits** shall be considered for this subject.
7. The students can do the MOOC according to their convenience but should **complete it by the fourth semester** of the M Plan (Housing) course.
8. Passing grades must be obtained before the last working day of the 4th semester for the award of credits without which the student shall not be eligible for award of M. Plan Degree.
9. The registration details for the MOOC as well as the completion certificate shall be submitted duly with the MOOC Coordinator of the institute.
10. Two credits are given for this on a pass or fail basis and it is mandatory for getting M Plan (Housing) degree. As no grade is given, this will not be used for CGPA calculation.

The suggested broad areas are defined as follows:

- **Environmental & Sustainable Planning** (Environmental impact assessment, green infrastructure, ecosystem services, circular economy, urban sustainability, compact city development, integrated waste management)
- **Urban Design, Landscape & Heritage** (Urban design, landscape planning, designing for real estate, public space design, urban form, nature-based design, urban heritage, conservation planning, adaptive reuse)
- **Housing, Real Estate & Land Systems** (Real estate development, housing finance, inclusive housing, slum rehabilitation, land value capture, TDR and ToD, land economics, peri-urban planning)
- **Infrastructure, Mobility & Transport Systems** (Urban mobility, intelligent transport systems, transport planning, urban logistics, last-mile connectivity, infrastructure policy, street design)
- **Governance, Policy & Legal Frameworks** (Urban governance, fiscal policy, legal studies, PPP, municipal finance, policy reforms, stakeholder management)
- **Data, Digital & Smart Planning Systems** (GIS, remote sensing, spatial data security, big data, AI in planning, smart cities, digital twins)
- **Disaster Risk, Resilience & Emergency Planning** (Disaster management, urban flood risk, resilience planning, emergency response, climate adaptation, recovery planning)
- **Economics, Policy, and Development Planning** (Urban economics, Development policy, Public finance, Environmental economics, Regional planning, Governance and participation)
- **Social Inclusion, Participation & Citizenship** (Community participation, inclusive planning, citizen engagement, behavioral governance, healthy communities)

* Students can also choose topics / courses relevant to Housing or Urban Planning offered by agencies, namely AICTE / NPTEL / SWAYAM with the recommendation of the Board of Studies (at least 70% of the course content shall be matched with the area/stream of study) and approval by the Academic Council.

*The Course shall not be considered if its content has more than 50% of overlap with a core/elective course in the concerned discipline or with an open elective.

Approved MOOCs providers:

SWAYAM MHRD <https://swayam.gov.in>

NPTEL IITs, IISc <https://onlinecourses.nptel.ac.in>

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